3.19 Fields and Galois Theory

3.19.1 Prime Fields

Recall that for p prime the quotient ring $\mathbb{Z}/p\mathbb{Z}$ is a field (3.16.3), (3.17.22), (3.3.59).

Definition 3.19.1 (Finite field of order p)
Denote by $\mathbb{F}_p$ the field $\mathbb{Z}/p\mathbb{Z}$ of order p.

Definition 3.19.2 (Rational Integers)
*We denote by $\mathbb{Q}$ the field of **rational numbers** defined to be the quotient field of the integers,* $\mathrm{Frac}(\mathbb{Z})$ *(see (3.6.6)).*

Definition 3.19.3 (Prime Field)
*We say that a field k is a **prime field** if it is isomorphic to one of $\mathbb{F}_p$ or $\mathbb{Q}$.*

Note none of these fields are mutually isomorphic by considering cardinalities.

Proposition 3.19.4 (Prime Subfield Exists)
*Let k be a field. Then k contains a **prime subfield**. It is the smallest subfield contained in k.*

Proof. By (...) there is a unique ring homomorphism

$$\phi : \mathbb{Z} \to k$$

As k is not the zero-ring then $\phi(1) = 1 \neq 0$ and $\ker(\phi)$ is a proper ideal. Then $\mathrm{Im}(\phi)$ is an integral domain by (3.3.9). By (3.3.57) $\mathbb{Z}/\ker(\phi) \cong \mathrm{Im}(\phi)$ and therefore by (3.3.59) $\ker(\phi) =: \mathfrak{p}$ is a prime ideal.

By (3.16.3) the ideal $\mathfrak{p}$ is principal. Suppose $\mathfrak{p} = (0)$. By (3.6.5) ϕ extends to a homomorphism $\phi' : \mathbb{Q} \to k$ whose kernel is zero and therefore injective (...). In particular $\mathrm{Im}(\phi')$ is a prime subfield being isomorphic to $\mathbb{Q}$.

Otherwise by (3.17.9) $\mathfrak{p} = (p)$ for p a prime number and we have already observed that $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \cong \mathrm{Im}(\phi)$ is a prime subfield.

Let k' be any subfield of k, then by induction we may show that $\mathrm{Im}(\phi) \subset k'$. In the case $\ker(\phi) = \{0\}$ then we may also show $\mathrm{Im}(\phi') \subset k'$. Therefore k' contains the prime subfield, whence it is the smallest subfield. ∎

Definition 3.19.5 (Characteristic)
*Let k be a field. Then we define the **characteristic** of k to be p if the prime subfield is isomorphic to $\mathbb{F}_p$ or 0 if the prime subfield is isomorphic to $\mathbb{Q}$. This is denoted* $\mathrm{char}(k)$.

*If A is a k-algebra then we define the characteristic of A to be that of k. We also define the **characteristic exponent** of A to be*

$$\begin{cases} 1 & \text{if } \mathrm{char}(A) = 0 \\ p & \text{if } \mathrm{char}(A) = p \end{cases}$$

Proposition 3.19.6 (Frobenius Homomorphism)
Let A be a k-algebra and suppose $\mathrm{char}(A) = p$. *Then the mapping*

$$a \to a^p$$

*is a ring homomorphism which we call the **Frobenius map**. In particular*

$$(a+b)^p = a^p + b^p \quad \forall a, b \in A$$

Definition 3.19.7
*Let A be a k-algebra with characteristic exponent p. We say that A is **perfect** if A is reduced and $A^p = A$. When $p > 1$ this is equivalent to the Frobenius homomorphism being an isomorphism.*

3.19.2 Separability

We consider separability of a polynomial, and how it behaves under field extension.

Definition 3.19.8 (Splitting Polynomial)
*Let K/k be a field extension and $f \in k[X]$. By abuse of notation we may also identify f with its image in $K[X]$. We say a polynomial f **splits completely** in K if the irreducible factorization of f in $K[X]$ is*

$$f = \ell(f) \prod_{i=1}^{n} (X - \alpha_i)^{r_i}$$

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where α_i are the distinct roots of f in K and $r_i := v_{X-\alpha_i}(f)$ are the multiplicities. Equivalently f splits in K if

$$p \in K[X] \text{ irreducible} \wedge \deg(p) > 1 \implies v_p(f) = 0$$

Observe that the number of roots counting multiplicities is $\deg(f)$

$$\deg(f) = \sum_{i=1}^{n} v_{X-\alpha_i}(f)$$

Corollary 3.19.9
Let K/k be a field extension and $f, g \in k[X]$. Suppose $g \mid f$ (in $k[X]$) and f splits completely in K. Then so does g.

Proof. As $k[X] \subset K[X]$, $g \mid f$ in $K[X]$ too. By assumption the irreducible factorization of f in $K[X]$ consists of polynomials of degree 1. Consider the irreducible factorization of $g = \prod_{i=1}^{n} g_i$ in $K[X]$, then by unique factorization (3.17.33) (applied to $K[X]$) each g_i must appear in the factorization of f, that is to say g splits completely in K. ■

Lemma 3.19.10 (Coprime Polynomials have no Common Roots)
Let K/k be a field extension and $f, g \in k[X]$ with $f \neq 0$ splitting completely in K. Then f and g are coprime if and only if they have no common root in K. In particular this holds whenever f and g both split completely in K.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose f, g are coprime, so $af + bg = 1$ for some $a, b \in k[X]$. If $\alpha \in K$ were a common root then evaluating at α gives $0 = 1$, a contradiction.

Conversely suppose f, g are not coprime. As $k[X]$ is a PID (3.17.32), by (3.16.5) they have a common non-invertible divisor d, and $\deg(d) > 0$ as the invertible elements of $k[X]$ are the nonzero constants (3.8.15). Since $d \mid f$ and f splits completely in K, so does d (3.19.9); as $\deg(d) > 0$ it has a root $\alpha \in K$, which is a common root of f and g as d divides both. ■

We can provide a partial converse to (3.17.39) by working in a large enough extension field

Proposition 3.19.11 (Separability)
Let K/k be a field extension and $f \in k[X]$ a polynomial which splits completely in K. Then TFAE

a) f is separable
b) f has no multiple roots in K
c) f and f' have no common roots in K
d) f has $\deg(f)$ distinct roots in K

Proof. Using the formula

$$\deg(f) = \sum_{i=1}^{n} v_{X-\alpha_i}(f)$$

we see easily that $d) \iff b)$. By (3.17.37) $b) \iff c)$. As f splits completely in K, (3.19.10) shows that f and f' are coprime, that is f is separable (3.17.38), if and only if they have no common root in K: $a) \iff c)$. ■

3.19.3 Field Extensions

Definition 3.19.12 (Field Extension)
Let k be a field. A field extension K/k is a k-algebra K which is also a field. Every field K is an extension over its prime subfield (or, over $\mathbb{Q}$ or $\mathbb{F}_p$).

We typically denote the structural morphism by $i_{kK} : k \to K$, and it is automatically injective (3.3.61). We may write $(K/k, i_{kK})$ if we need to stress the relevance of the structural morphism to the argument.

These objects form a category $\mathbf{Field}_k$ in the obvious way. The morphisms may be called k-embeddings and we denote them by

$$\mathrm{Mor}_k(K, L) := \{\psi : K \to L \mid \psi \circ i_{kK} = i_{kL}\}$$

and the set of automorphisms by

$$\mathrm{Aut}(K/k).$$

Observe every extension K/k may be viewed as a k-vector space so we define the degree of an extension field to be the vector space dimension

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$$[K:k] := \dim_k K$$

Definition 3.19.13 (Finite field extension)
A field extension K/k is finite if $[K:k] < \infty$

Definition 3.19.14 (Tower of Field Extensions)
We may also consider a "tower" of extensions

$$K_n/\cdots/K_0 = k$$

with embeddings $i_{K_iK_{i+1}} : K_i \to K_{i+1}$, with the picture that these usually correspond to inclusions. We may consider an extension K_i/K_j for $j < i$. Typically if we have a family of morphisms

$$\sigma_i : K_i \to M$$

they would commute with these embeddings. In particular we may abuse notation by defining $\sigma_i|_{K_j} = \sigma_i \circ i_{K_{i-1}K_i} \circ \cdots \circ i_{K_jK_{j+1}}$.

Proposition 3.19.15
Let K/k be a finite field extension and V a finite-dimensional vector space over K. Then V is naturally a finite-dimensional k-vector space with

$$\dim_k V = [K:k] \dim_K V$$

More precisely if K/k has basis $\{\lambda_1, ..., \lambda_r\}$ and V has K-basis $\{v_1, ..., v_n\}$, then the following is a k-basis for V

$$\left\{\lambda_i v_j\right\}_{i,j}.$$

Proposition 3.19.16
Let L/K and K/k be two finite extensions with basis $\{l_1, ..., l_n\}$ and $\{k_1, ..., k_m\}$. Then L/k has basis $\left\{l_i k_j\right\}_{i,j}$. In particular

$$[L:k] = [L:K][K:k]$$

Corollary 3.19.17
Let $K = K_n/\cdots/K_0 = k$ be a tower of finite extensions then

$$[K:k] = \prod_{i=1}^{n} [K_i : K_{i-1}]$$

Lemma 3.19.18
Let K/k be a field extension and $f, g \in k[X]$. Then $g \mid f$ in $k[X]$ if and only if $i_{kK}(g) \mid i_{kK}(f)$.

Proof. One implication is obvious. For the converse we assume wlog that $k \subset K$ and suppose $f = gh$ for $h \in K[X]$. We may apply the division algorithm (3.8.16) in $k[X]$ to find $f = gq + r$ for $q, r \in k[X]$ and $\deg(r) < \deg(g)$. Then $r = g(h - q)$ and comparing degrees we conclude that $h = q$ and $r = 0$. This shows $f = gq$ and $g \mid f$ as elements of $k[X]$. ∎

Definition 3.19.19 (Evaluation homomorphism)
Let K/k be a field extension and $\alpha \in K$. There is a canonical homomorphism

$$\mathrm{ev}_\alpha : k[X] \to K$$

$$\sum_{i=0}^{n} a_i X^i \to \sum_{i=0}^{n} i_{kK}(a_i)\alpha^i$$

which we write as $f(\alpha)$. We say $\alpha \in K$ is a root of $f(X)$ if $f(\alpha) = 0$.

Proposition 3.19.20 (Morphisms commute with evaluation)
Let $\sigma : K/k \to L/k$ be a morphism of field extensions then

$$\sigma(p(\alpha)) = p(\sigma(\alpha))$$

for all $p \in k[X]$. In particular α is a root of $p \iff \sigma(\alpha)$ is a root of p.

Proof. This is just a specific case of (3.8.8). The last statement is obvious, because σ is injective (3.3.61). ∎

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Definition 3.19.21 (Subalgebra generated by a set)
Let K/k be a field extension and $S \subset K$. Define

$$k[S] := \bigcap_{\substack{A \subset K/k \\ S \subset A}} A$$

where the intersection is taken over all k-subalgebras. It is the smallest k-subalgebra of K/k.

Proposition 3.19.22 (Subalgebra generated by directed family)
Let K/k be a field extension and $\{S_i\}_{i \in I}$ a family of directed subsets of K. Then

$$k\left[\bigcup_{i \in I} S_i\right] = \bigcup_{i \in I} k[S_i]$$

In particular for any set S we have

$$k[S] = \bigcup_{\substack{S' \subset S \\ S' \text{ finite}}} k[S']$$

Proposition 3.19.23 (Subalgebra generated by a set)
Let K/k be a field extension and $S \subset K$ a finite subset. Write $S = \{\alpha_1, ..., \alpha_n\}$ then

$$k[S] = \{p(\alpha_1, ..., \alpha_n) \mid p \in k[X_1, ..., X_n]\} = \mathrm{Im}(\mathrm{ev}_\alpha)$$

Lemma 3.19.24 (Trivial result)
For $S, T \subset K/k$ finite

- $S \subset T \implies k[S] \subseteq k[T]$
- $k[S][T] = k[S \cup T]$

Definition 3.19.25
Let K/k be a field extension and $S \subset K$. Define

$$k(S) := \bigcap_{\substack{K' \subset K/k \\ S \subset K'}} K'$$

It is the smallest subfield of K/k containing S.

Proposition 3.19.26
Let K/k be a field extension and $\{S_i\}_{i \in I}$ be a family of directed subsets of K. Then

$$k\left(\bigcup_{i \in I} S_i\right) = \bigcup_{i \in I} k(S_i)$$

In particular for any set S we have

$$k(S) = \bigcup_{\substack{S' \subset S \\ \text{finite}}} k(S')$$

Proposition 3.19.27
Let K/k be a field extension and $S \subset K$ be a finite subset. Write $S = \{\alpha_1, ..., \alpha_n\}$ then

$$k(S) := \left\{ \frac{p(\alpha_1, ..., \alpha_n)}{q(\alpha_1, ..., \alpha_n)} \;\middle|\; p, q \in k[X_1, ..., X_n], \quad q(\alpha_1, ..., \alpha_n) \neq 0 \right\}$$

Lemma 3.19.28 (Trivial result)
For $S, T \subset K/k$

- $S \subset T \implies k(S) \subseteq k(T)$
- $k(S)(T) = k(S \cup T)$

Lemma 3.19.29
If $S \subset K$ and $k[S]$ is a field then $k[S] = k(S)$

Proof. Generically $k[S] \subset k(S)$ from the definition. As $k[S]$ is a field then $k(S) \subset k[S]$. ∎

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Lemma 3.19.30 (Image of f.g. field extension)
Let K/k be a field extension and $S \subset K$ a subset. If $\sigma : K/k \to L/k$ is a morphism them

$$\sigma(k(S)) = k(\sigma(S))$$

Proposition 3.19.31 (Uniqueness of morphisms on a generating set)
Let K/k be a field extension and $S \subset K$. If $\sigma, \sigma' : k(S)/k \to L/k$ are morphisms of field extensions such that $\sigma|_S = \sigma'|_S$. Then $\sigma = \sigma'$.

Definition 3.19.32 (Simple (Algebraic) Extension)
*A field extension K/k is **simple** if $K = k(\{\alpha\}) =: k(\alpha)$ for some $\alpha \in K$. It is a **simple algebraic** extension if α is also algebraic over k.*

Definition 3.19.33 (Algebraic Element)
*We say an element $\alpha \in K/k$ is **algebraic** if it is a root of a polynomial $f \in k[X]$ (i.e. α is integral, since we can always ensure f is monic). Otherwise we say that $x \in K$ is **transcendental**.*

*We say K/k is an **algebraic extension** if every element $\alpha \in K$ is algebraic over k.*

Proposition 3.19.34 (Finite ⇒ algebraic)
A finite extension K/k is algebraic.

Proof. Suppose $n = \dim_k K$. The set $\{1, \alpha, \alpha^2, ..., \alpha^n\}$ is linearly dependent by (2.3.13). Therefore there is a non-zero polynomial with α as a root. ∎

Proposition 3.19.35 (Endomorphisms are automorphisms)
Let $\sigma \in \mathrm{Mor}_k(K, K)$ be an endomorphism of an algebraic extension. Then it is an isomorphism. In other words

$$\mathrm{Mor}_k(K, K) = \mathrm{Aut}(K/k)$$

Proof. As field morphisms are injective (3.3.61) we only need to show that σ is surjective. Given $\alpha \in K$ let T denote the set of roots of $m_\alpha \in k[X]$ in K. Note by (3.17.35) T is finite. Further by (3.19.20) σ maps T to itself. Since σ is injective it is also surjective on T. In particular α is in the image of σ as required. ∎

3.19.4 Algebraic Extensions

Proposition 3.19.36 (Minimal Polynomial)
If $\alpha \in K/k$ is algebraic then there is a unique monic, irreducible polynomial $m_{\alpha,k}(X) \in k[X]$ such that $m_{\alpha,k}(\alpha) = 0$. This is called the minimal polynomial of α over k and $\big(m_{\alpha,k}\big) = \ker(\mathrm{ev}_\alpha)$.

In particular any polynomial $f(X) \in k[X]$ which has α as a root, satisfies $m_{\alpha,k}(X) \mid f(X)$.

Proof. Let $\mathfrak{a} = \ker(\mathrm{ev}_\alpha)$. Since $k[X]$ is a PID it is of the form $\big(m_{\alpha,k}\big)$. As α is algebraic it is non-zero. $m_{\alpha,k}(X)$ cannot be a constant, and therefore is not a unit.

We claim $m_{\alpha,k}$ is irreducible. If $m_{\alpha,k}(X) = p(X)q(X)$ then p, q are non-zero and either $p(\alpha) = 0$ or $q(\alpha) = 0$. If $p(\alpha) = 0$ then $m_{\alpha,k} \mid p$. As $p \mid m_{\alpha,k}$ by (3.17.30) $m_{\alpha,k} = \lambda p$. In particular $\deg\big(m_{\alpha,k}(X)\big) = \deg(p(X))$ so $\deg(q(X)) = 0$ and $q(X)$ is a unit (3.8.15). Therefore by definition $m_{\alpha,k}(X)$ is irreducible.

Dividing by the leading coefficient we may assume that this polynomial is monic. Suppose $m'(X)$ is another such irreducible monic polynomial. Then $m_\alpha \mid m'$. Since m_α is not a unit, by definition of irreducible $m' \sim m_\alpha$ whence $m' = \lambda m_\alpha$. Compare leading coefficients to find $\lambda = 1$ and $m' = m_\alpha$. ∎

Lemma 3.19.37
Let $K/E/k$ be extensions and $\alpha \in K$ algebraic over k. Then α is algebraic over E.

Definition 3.19.38 (Conjugate elements)
*Two elements $\alpha, \beta \in K$ are said to be **conjugate elements** if they have the same minimal polynomial.*

NB it's necessary and sufficient that $m_{\alpha,k}(\beta) = 0$.

Proposition 3.19.39
Let $\sigma : K/k \to L/k$ be a field morphism and $\alpha \in K$. Then $m_{\alpha,k}(X) = m_{\sigma(\alpha),k}(X)$.

Proof. This follows from (3.19.20). ∎

Given an irreducible polynomial $f \in k[X]$ it's possible to construct an extension field K/k which has at least one root, as follows.

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Proposition 3.19.40 (Construct simple extension)
Let $f \in k[X]$ be an irreducible polynomial. Then (f) is maximal and $K := k[X]/(f)$ is a field extension with canonical structural morphism. Define $\alpha := X + (f)$

- $f(\alpha) = 0$
- $K = k(\alpha)$ *is a simple field extension and* $k(\alpha) = k[\alpha]$
- $m_\alpha = f/\ell(f)$ *and* $\deg(m_\alpha) = \deg(f) =: n$
- K *is a finite-dimensional k-vector space with basis*

$$\{1, \alpha, ..., \alpha^{n-1}\}$$

.

Example 3.19.41
Take $k = \mathbb{R}$, $f(X) = X^2 + 1$, then $\mathbb{C}/\mathbb{R} = \mathbb{R}[i] = \mathbb{R}[X]/(X^2 + 1)$.

Proof. Consider the structural morphism $i : k \to k[X]$ and canonical surjective homomorphism

$$\pi : k[X] \to k[X]/(f)$$

and $\alpha = X + (f) = \pi(X)$. As $k[X]$ is a PID, f irreducible implies (f) maximal by (3.17.22) so K is a field by (3.3.59). The composition $\pi \circ i$ makes K into a k-algebra and hence a field extension. Furthermore π is then by definition a k -algebra homomorphism.

Since π is surjective every $\beta \in K$ is represented as $\pi(p(X)) \overset{(3.19.20)}{=} p(\pi(X)) = p(\alpha)$. By (3.19.23) we see $K = k[\alpha]$. Since K is a field then $K = k[\alpha] = k(\alpha)$ is simple by (3.19.29).

Similarly $f(\alpha) = f(\pi(X)) \overset{(3.19.20)}{=} \pi(f(X)) = 0$, so α is a root of f. By (3.17.30) $f/\ell(f)$ is irreducible and by uniqueness in (3.19.36) we have $m_\alpha = f/\ell(f)$.

Given $\beta = p(\alpha)$, the division algorithm (...) yields

$$p(X) = q(X)f(X) + r(X)$$

with $\deg(r) < \deg(f) = n$. Therefore $\beta = r(\alpha)$ and the given set is spanning. A non-trivial linear dependence yields a non-zero polynomial $g(X)$ such that $g(\alpha) = 0$ and $\deg(g) < \deg(f)$. But by definition of the minimal polynomial $m_\alpha \mid g$, which is a contradiction by comparing degrees. Therefore the given set is linearly independent and hence a basis. ∎

Conversely any simple algebraic extension is obtained in this way, as follows

Proposition 3.19.42 (Simple extension)
Let $k(\alpha)/k$ be a simple extension. Then there is a canonical isomorphism of k-algebras

$$k[X]/(m_\alpha) \longrightarrow k(\alpha)$$

under which $X + (m_\alpha) \to \alpha$. Further $k(\alpha)$ is a finite-dimensional vector space with basis

$$\{1, \alpha, ..., \alpha^{n-1}\}$$

where $n = \deg(m_\alpha) = [k(\alpha) : k]$ and $k(\alpha) = k[\alpha]$.

Proof. By (3.19.36), Definition (3.19.23) and (3.3.57) there is a canonical isomorphism $k[X]/(m_\alpha) \to k[\alpha]$ of k-algebras induced by the evaluation homomorphism $\mathrm{ev}_\alpha : k[X] \to K$. (3.19.40) shows that the image of this isomorphism, $k[\alpha]$, is a field, whence $k[\alpha] = k(\alpha)$ by (3.19.29). Since a k-algebra isomorphism is a-fortiori a k-vector space isomorphism it maps a basis to a basis. The result follows from (3.19.40) as the basis thus defined is the image of the basis in the proposition under the specified isomorphism. ∎

Definition 3.19.43 (Degree of an algebraic element)
Let K/k be an algebraic extension and $\alpha \in K$. Then define

$$\deg_k(\alpha) := \deg m_{\alpha,k} = [k(\alpha) : k]$$

We may show the following

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Proposition 3.19.44 (Finitely generated by algebraic $\implies$ finite and algebraic)
Let $K = k(\alpha_1, ..., \alpha_n)/k$ be a field extension such that α_i is algebraic. Then K/k is a finite algebraic extension. Furthermore

$$k[\alpha_1, ..., \alpha_n] = k(\alpha_1, ..., \alpha_n)$$

In particular a finitely-generated algebraic extension is finite.

Proof. We write $K_i = k(\alpha_1, ..., \alpha_i)$. Then we have a tower

$$K = K_n/.../K_0 = k$$

such that $K_i = K_{i-1}(\alpha_i)$ is a simple algebraic extension. By (3.19.42) K_i/K_{i-1} is finite. Therefore by (3.19.17) K/k is finite. By (3.19.34) it's also algebraic. For the second statement we may proceed inductively. Note we have

$$k[\alpha_1, ..., \alpha_{i+1}] \overset{(3.19.24)}{=} k[\alpha_1, ..., \alpha_i][\alpha_{i+1}] = k(\alpha_1, ..., \alpha_i)[\alpha_{i+1}] \overset{(3.19.42)}{=} k(\alpha_1, ..., \alpha_i)(\alpha_{i+1}) \overset{(3.19.28)}{=} k(\alpha_1, ..., \alpha_{i+1})$$

The second equality is simply the inductive hypothesis. ∎

Corollary 3.19.45
Let K/k be a field extension then the algebraic elements form a subfield.

Proof. For any two algebraic elements $\alpha, \beta \in K$ we have $k(\alpha, \beta)$ is an algebraic extension. ∎

The following is useful for reducing to cases of finite extensions where counting arguments work.

Lemma 3.19.46 (Reduce to finite extensions)
Let $K/E/k$ be a tower with E algebraic over k. For every $\alpha \in K$ algebraic over E, there is some subfield $E_0 \subset E$ such that

- *E_0/k is finite*
- *α is algebraic over E_0*
- *$m_{\alpha,E} = m_{\alpha,E_0}$*

Therefore α is algebraic over k iff it is algebraic over E and $m_{\alpha,E_0} \mid m_{\alpha,k}^{i_{kE}}$.

Proof. Suppose

$$m_{\alpha,E}(X) = a_0 + a_1 X + ...a_n X^n$$

Then define $E_0 = i_{kE}(k)(a_0, ..., a_n)$. By (3.19.44) E_0/k is finite. Clearly α is algebraic over E_0 as it is a root of $m_{\alpha,E}$. By (3.19.36) $m_{\alpha,E_0} \mid m_{\alpha,E}$ as elements of $E_0[X]$ and $m_{\alpha,E} \mid m_{\alpha,E_0}$. Therefore $m_{\alpha,E_0} = m_{\alpha,E}$.

By (3.19.42) $E_0(\alpha)/E$ is finite, therefore $E_0(\alpha)/k$ is finite. By (3.19.34) $E_0(\alpha)/k$ is algebraic, whence α is algebraic over k. The last statement follows from (3.19.36) again. ∎

Corollary 3.19.47 (Algebraic Extensions are Directed Unions of Finite Subextensions)
Let E/k be algebraic. Then the finite subextensions of E/k are directed under inclusion, and E is their union.

Proof. Given $\alpha \in E$, taking $K = E$ in (3.19.46) (with α trivially algebraic over E) gives a finite subextension $E_0 \subset E$ over which α is algebraic; then $E_0(\alpha)/E_0$ is finite (3.19.42), hence $E_0(\alpha)/k$ is finite too, a finite subextension of E containing α. Hence E is the union of its finite subextensions. Given two finite subextensions $E_1, E_2 \subset E$, their compositum $E_1E_2 \subset E$ is finitely-generated and algebraic over k, hence finite (3.19.44), and contains both, so the finite subextensions are directed under inclusion. ∎

Corollary 3.19.48
K/E and E/k are both algebraic if and only if K/k is.

Proof. One direction is (3.19.37). The converse follows from the previous result. ∎

We may prove the first lifting theorem

Proposition 3.19.49 (Lifting to simple extensions)
Let $k(\alpha)/k$ be a simple algebraic extension and L/k a field extension such that $m_{\alpha,k}$ has a root in L. Then there exists a morphism $\sigma : k(\alpha)/k \to L/k$.

More precisely there is a bijective mapping

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$$\begin{aligned}\mathrm{Mor}_k(k(\alpha), L) &\longrightarrow \{\beta \in L \mid m_{\alpha,k}(\beta) = 0\} \\ \sigma &\to \quad \sigma(\alpha)\end{aligned}$$

and $\sigma(k(\alpha)) = k(\sigma(\alpha))$*. In particular if* $m_{\alpha,k}$ *is separable and splits completely in* L *then there are precisely* $\deg(m_\alpha) \overset{(3.19.42)}{=} [k(\alpha) : k]$ *such extensions.*

Proof. Observe $m_{\alpha,k}(\sigma(\alpha)) \overset{(3.19.20)}{=} \sigma\big(m_{\alpha,k}(\alpha)\big) = 0$. Therefore the mapping is well-defined. By (3.19.31) it is injective. We claim it is also surjective. By (3.19.42) there is a k-algebra isomorphism

$$k[X]/\big(m_{\alpha,k}\big) \longrightarrow k(\alpha)$$

Similarly for $\beta \in T$ there is a k-algebra isomorphism

$$k[X]/\big(m_{\beta,k}\big) \longrightarrow k(\beta)$$

We are done if $m_{\alpha,k} = m_{\beta,k}$. But $m_{\alpha,k}$ is monic, irreducible and has β as a root. So this follows from uniqueness of the minimal polynomial in (3.19.36). The final statement follows from (3.19.11) ∎

We may use this to generalize to arbitrary extensions, but we require that the minimal polynomials split completely in order for the inductive step to work.

Proposition 3.19.50 (Generic Lifting Theorem)
Let K/k *be an algebraic field extension such that* $K = k\big(\{\alpha_i\}_{i\in I}\big)$ *and* L/k *a field extension such that* $m_{\alpha_i,k}(X)$ *splits completely in* L *for all* $i \in I$*.*

Then there exists a morphism $\sigma : K/k \to L/k$*.*

Furthermore given $\alpha \in K$ *and* $\beta \in L$ *any root of* $m_{\alpha,k}(X)$ *we may choose* σ *such that* $\sigma(\alpha) = \beta$*.*

Proof. If K/k is finite then we may proceed by induction on $[K : k]$, using (3.19.49) and applying a similar argument to below.

For the general case we may consider the poset of morphisms $\sigma : K'/k \to L/k$ for subfields $K'/k \subset K/k$ ordered by consistency. It is non-empty by considering $K' = i_{kK}(k)$. By Zorn's Lemma it has a maximal element, (K', σ'). It's enough to show that $K' = K$.

If $\alpha_i \in K'$ for all $i \in I$ then $K' = K$ and we are done. Otherwise choose $\alpha = \alpha_i \notin K'$. By (...) $m_{\alpha,K'}(X) \mid m_{\alpha,k}(X)$. By (3.19.9) $m_{\alpha,K'}(X)$ splits in L (because $m_{\alpha,k}(X)$ does). Therefore by (3.19.49) there is a morphism $\sigma : K'(\alpha)/K' \to (L/K', \sigma')$. Note that by definition $\sigma|_{K'} = \sigma'$ and $K' \subsetneq K'(\alpha)$, contradicting maximality.

For the final part we may consider the poset consisting only of morphisms such that $\sigma(\alpha) = \beta$. By (3.19.49) the poset is non-empty, and the same argument works. ∎

3.19.5 Galois Theory Summary

Definition 3.19.51 (Separable, Normal and Galois)
Let K/k *be an algebraic extension. We say that* K/k *is*

- ***Normal*** *if every minimal polynomial* $m_{\alpha,k} \in k[X]$ *splits completely in* K *(iff every irreducible polynomial* $f \in k[X]$ *with at least one root in* K *splits completely in* K*)*
- ***Separable*** *if every minimal polynomial* $m_{\alpha,k} \in k[X]$ *is separable.*
- ***Galois*** *if it is both normal and separable (iff* $m_{\alpha,k}$ *has* $\deg\big(m_{\alpha,k}\big)$ *distinct roots in* K*, see (3.19.11)).*

In the case of a Galois extension we denote the group of automorphisms by $\mathrm{Gal}(K/k)$*.*

To summarize the main results

a) The group of automorphism of a normal extension K/k acts transitively on the roots of a given irreducible polynomial.
b) For K/k finite we have $\#\,\mathrm{Aut}(K/k) \leq [K : k]$ with equality if and only if K/k is Galois.
c) An algebraic extension K/k is automatically separable whenever either $\mathrm{char}(k) = 0$ or k is finite.
d) When K/k is finite and Galois then we have an order-reversing bijection between subfields and subgroups

$$\begin{aligned}\{H \leq \mathrm{Gal}(K/k)\} &\longleftrightarrow \{F \subseteq K\} \\ H &\longrightarrow K^H := \{x \in K \mid h(x) = x \quad \forall h \in H\} \\ \mathrm{Gal}(K/F) &\longleftarrow F\end{aligned}$$

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3.19.6 Splitting Fields and Algebraic Closure

In this section we discuss splitting fields, which are the "smallest" extensions in which a given set of polynomials split completely. The fundamental result is that splitting fields are precisely the Normal extensions. Further we discuss the algebraic closure, in which every polynomial splits and in which every algebraic extension (normal or otherwise) may be embedded.

Definition 3.19.52 (Splitting field)
*Let $S \subset k[X]$ a family of polynomials. We say that K/k is a **splitting field** for S if*

- *Every polynomial $f \in S$ splits completely in K*
- *K is generated by the roots of all the polynomials in S*

Note that by (3.19.44) K/k is necessarily algebraic, and if S is finite then so is K/k.

Definition 3.19.53 (Set of roots)
Let K/k be a field extension and $f \in k[X]$. Then define

$$T_{f,K} \coloneqq \{\beta \in K \mid f(\beta) = 0\}$$

Proposition 3.19.54 (Splitting field is minimal)
Let $S \subset k[X]$ be a family of polynomials which split completely in K/k. Then the following are equivalent

- *K is generated by the roots of every polynomial $f \in S$, that is*

$$K = k\left(\bigcup_{f\in S} T_{f,K}\right)$$

- *Any subfield $K' \subset K$ in which S splits completely is equal to K*

Proof. For all $f \in S$ there is a factorisation

$$f = \prod_{\alpha \in T_{f_i,K}} (X - \alpha)$$

Suppose $K = k\left(\bigcup_{f\in S} T_{f,K}\right)$ and let K' be a subfield in which all $f \in S$ split completely. Then by unique factorization in $K[X]$ we have $T_{f,K} \subset K'$ for all $f \in S$ and therefore $K' = K$.

Conversely it's clear that S splits completely in $k\left(\bigcup_{f\in S} T_{f,K}\right)$, therefore by hypothesis this equals K. ∎

Lemma 3.19.55
Let $\sigma : K/k \to L/k$ be a morphism and $f(X) \in k[X]$ a polynomial. Then

- *σ induces an injective map on the roots $T_{f,K} \to T_{f,L}$*
- *f splits completely in K iff f splits completely in $\sigma(K)$. In this case the above map is a bijection*

Proposition 3.19.56 (Image of a splitting field is fixed)
Let K/k be a splitting field for S and $\sigma : K/k \to L/k$ a morphism. Then S splits completely in L. Any such σ satisfies

$$\sigma(K) = k\left(\bigcup_{f\in S} T_{f,L}\right)$$

Proof. Clearly by (3.19.55) S splits completely in L.

By the same result σ induces a bijection $T_{f,K} \longleftrightarrow T_{f,L}$. Therefore $\sigma(K) = \sigma\left(k\left(\bigcup_{f\in S} T_{f,K}\right)\right) = k\left(\sigma\left(\bigcup_{f\in S} T_{f,K}\right)\right) = k\left(\bigcup_{f\in S} T_{f,L}\right)$ by (3.19.30). ∎

Proposition 3.19.57 (Uniqueness of Splitting Fields)
Let $S \subset k[X]$ be a family of polynomials. Let K/k be a splitting field for S and L/k an extension in which S splits completely.

Then there exists a morphism $\sigma : K/k \to L/k$. Let $\alpha \in K$ and $\beta \in L$ be conjugate elements, then we may choose σ such that $\sigma(\alpha) = \beta$.

Furthermore any two splitting fields are isomorphic.

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Proof. By assumption K is generated by the roots α_{ij} of $f_i \in S$. For each α_{ij} we therefore have $m_{\alpha_{ij},k}(X) \mid f_i(X)$ and $m_{\alpha_{ij},k}(X)$ splits completely in L by (3.19.9). Therefore the morphism $\sigma : K/k \to L/k$ exists by (3.19.50).

Note by (3.19.56) S splits in $\sigma(K) = k\left(\bigcup_{f\in S} T_{f,L}\right)$. If L is also a splitting field for S then $L = \sigma(K)$ by (3.19.54) and therefore σ is an isomorphism as required. ∎

Proposition 3.19.58 (Algebraically Closed)
A field M is algebraically closed if one of the following equivalent conditions holds

- *Every algebraic extension M'/M is trivial*
- *Every non-constant polynomial in $M[X]$ has a root in M*
- *Every non-constant polynomial in $M[X]$ splits in M*

NB in this case M is also normal.

Definition 3.19.59 (Algebraic Closure)
*An **algebraic closure** $\overline{k}$ of k is a field extension $\overline{k}/k$ which is algebraic and for which $\overline{k}$ is algebraically closed.*

Proposition 3.19.60 (Existence of Algebraic Closure)
Given a field k there exists an algebraic closure $\overline{k}/k$

Proposition 3.19.61 (Algebraic extensions embed into Algebraic Closure)
Let K/k be an algebraic extension and M/k be field such that every polynomial $f \in k[X]$ splits completely then there exists a morphism $\sigma : K/k \to M/k$.

In particular this holds when M is algebraically closed.

Proof. A straightforward application of (3.19.50) since every $m_{\alpha,k}(X)$ splits in M. ∎

Corollary 3.19.62 (Uniqueness of algebraic closure)
An algebraic closure $\overline{k}$ of k is unique up to (non-unique) isomorphism.

More generally we may show the existence of smaller splitting fields

Proposition 3.19.63 (Existence of Splitting Field)
Given a field k and family of polynomials $S \subset k[X]$ then there exists a splitting field K.

When $S = \{f\}$ then this can be chosen such that $[K : k] \leq n!$ where $n = \deg(f)$.

Proof. We may take the subfield of $\overline{k}$ generated by the roots of polynomials in S.

In the case S is finite it is possible to avoid the use of $\overline{k}$. First reduce to the case of a single polynomial $S = \{f\}$ and proceed by induction on $\deg(f)$. The inductive step may be demonstrated using (3.19.40). ∎

Remark 3.19.64
Note if K/k is an algebraic extension then (3.19.61) shows that we may construct an embedding $K \to \overline{k}$ commuting with $k \to \overline{k}$.

In general given a tower of algebraic extensions

$$K = k_n/.../k_0 = k$$

we will assume the existence of compatible embeddings $i_{k_i} : k_i \to \overline{k}$ such that $i_{k_{i+1}} \circ i_{k_i,k_{i+1}} = i_{k_i}$.

3.19.7 Normal Extensions

Recall that an algebraic extension K/k is normal if all minimal polynomials split completely. They are in some sense "closed". Furthermore $\overline{k}/k$ is clearly normal and results about $\overline{k}$ can often be generalized to normal fields L/k. We also show that an extension is normal iff it is a splitting field.

Lemma 3.19.65
Let $L/K/k$ be a tower of algebraic extensions and $\alpha \in L$. If $m_{\alpha,k}(X)$ splits completely in L so does $m_{\alpha,K}(X)$. In particular

$$L/k \text{ normal} \implies L/K \text{ normal}$$

Proof. Note $m_{\alpha,K}(X) \mid m_{\alpha,k}^{i_{kK}}(X)$ as elements of $K[X]$ by (3.19.36). Apply i_{KL} and then we may use (3.19.9). ∎

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Proposition 3.19.66 (Conjugate elements in Normal Extensions)
Let L/k be a normal *extension (e.g. $L = \overline{k}$) and $\alpha, \beta \in L$ elements with the same minimal polynomial $m_\alpha(X) = m_\beta(X)$. Then there exists $\sigma \in \mathrm{Aut}(L/k)$ such that*

$$\sigma(\alpha) = \beta$$

Proof. Apply (3.19.50) with $K = L$. ∎

Proposition 3.19.67 (Normal Criteria)
Let $L/K/k$ be a tower of extensions such that L/k is normal (e.g. $L = \overline{k}$). Then the following are equivalent

- ***NOR1**: For any $\sigma \in \mathrm{Mor}_k(K, L)$ we have $\sigma(K) = i_{KL}(K)$.*
- ***NOR2**: K/k is the splitting field of some family of polynomials $f_i \in k[X]$.*
- ***NOR3**: K/k is normal*

Proof. Clearly $3 \implies 2$, for K is the splitting field of all the minimal polynomials of elements in K.

$2 \implies 1$. This is (3.19.56).

$1 \implies 3$. Consider any $\alpha \in K$ with minimal polynomial $m_{\alpha,k}(X)$. By definition $m_{\alpha,k}(X)$ splits completely in L because it has a root $\alpha_1 = i_{KL}(\alpha)$. Denote the roots by $\alpha_1, ..., \alpha_r$. By (3.19.66) there is $\sigma_j \in \mathrm{Aut}(L/k)$ such that $\sigma_j(\alpha_1) = \alpha_j$. By hypothesis we have $\alpha_j \in (\sigma_j \circ i_{KL})(K) = i_{KL}(K)$ whence there exists $\alpha_j' \in K$ such that $i_{KL}(\alpha_j') = \alpha_j$. By (3.19.55) $m_{\alpha,k}(X)$ splits completely in K. Therefore K/k is normal as required. ∎

Corollary 3.19.68 (Splitting fields are normal)
An algebraic extension K/k is normal if and only if it is a splitting field.

Proof. We may apply the previous Proposition with $L = \overline{k}$.

We may prove a splitting field is normal more directly (without recourse to $\overline{k}$ or Zorn's Lemma in the finite case). Suppose K/k is a splitting field for $S \subset k[X]$. Consider $\alpha \in K$ with minimal polynomial $m_{\alpha,k}(X)$. Let $(L/K, i_{KL})$ be a splitting field for $m_{\alpha,k}(X)$ (as a polynomial in $K[X]$, NB may not be irreducible).

Let $\beta \in L$ be another root of $m_{\alpha,k}(X)$. Observe that S splits in L, so by (3.19.57) there is a morphism $\sigma : K/k \to L/k$ with $\sigma(\alpha) = \beta$. By (3.19.56) we have $i_{KL}(K) = \sigma(K)$ whence $\beta \in i_{KL}(K)$. As β was an arbitrary root of $m_{\alpha,k}(X)$ we see it splits completely in $i_{KL}(K)$. Finally by (3.19.55) $m_{\alpha,k}(X)$ splits completely in K. As α was arbitrary then K/k is normal. ∎

Proposition 3.19.69 (Extension to normal overfield)
Let $L/K/k$ be a tower of algebraic field extensions with L/k normal (e.g. $L = \overline{k}$) then there is a canonical surjection

$$\begin{aligned} \mathrm{Mor}_k(i_{KL}, L) : \mathrm{Aut}(L/k) &\to \mathrm{Mor}_k(K, L) \\ \sigma &\to \sigma \circ i_{KL} \end{aligned}$$

When i_{KL} is inclusion then this is simply the restriction to K. The kernel is precisely $\mathrm{Aut}(L/K)$.

Proof. Given $\tilde{\sigma} \in \mathrm{Mor}_k(K, L)$, apply (3.19.50) to construct a morphism $\sigma : (L/K, i_{KL}) \to (L/K, \tilde{\sigma})$. The hypotheses apply because the minimal polynomial $m_{\alpha,K}(X)$ with respect to either extension divides the minimal polynomial $m_{\alpha,k}^{i_{kK}}(X)$ which by assumption splits completely in L. By (3.19.35) it is an automorphism. Furthermore

$$\sigma \circ i_{kL} = \sigma \circ i_{KL} \circ i_{kK} = \tilde{\sigma} \circ i_{kK} = i_{kL}$$

whence $\sigma \in \mathrm{Aut}(L/k)$ as required. ∎

Corollary 3.19.70 (Lifting inside normal overfield)
Let $L/K/F/k$ be a tower of field extensions with L/k normal, then there is a surjection

$$\mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(F, L)$$

Proof. Note that $\mathrm{Mor}_k(i_{FK}, L) \circ \mathrm{Mor}_k(i_{KL}, L) = \mathrm{Mor}_k(i_{FL}, L)$. By (3.19.69) this composition is surjective, whence the result follows. ∎

Corollary 3.19.71 (Quotient of automorphism group)
Let $L/K/k$ be a tower of extensions such that L/k and K/k are normal. Then L/K is normal and there is an isomorphism of groups.

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$$\begin{aligned} \operatorname{Aut}(L/k)/\operatorname{Aut}(L/K) &\longrightarrow \operatorname{Aut}(K/k) \\ \sigma &\to \quad i_{KL}^{-1} \circ \sigma \circ i_{KL} \end{aligned}$$

Proof. The given map is well-defined by (3.19.67) since $(\sigma \circ i_{KL})(K) = i_{KL}(K)$, and σ fixes K precisely when $\sigma \circ i_{KL} = i_{KL}$, that is $\sigma \in \operatorname{Aut}(L/K)$. Given $\tau \in \operatorname{Aut}(K/k)$, by (3.19.69) there exists $\sigma \in \operatorname{Aut}(L/k)$ such that $\sigma \circ i_{KL} = i_{KL} \circ \tau$. This shows the given map is surjective. The result follows from the group isomorphism theorem. ∎

Definition 3.19.72 (Normal Closure)
*Let K/k be an algebraic extension. Then an algebraic extension L/K is a **normal closure** for K/k if*

- *L/k is normal*
- *No proper subfield $i_{KL}(K) \subseteq L' \subsetneq L$ is normal over k*

Proposition 3.19.73 (Existence and Uniqueness of Normal Closure)
Let K/k be an algebraic extension. Then a normal closure L/K exists and is unique up to isomorphism. Indeed it is the splitting field for all the minimal polynomials $\{m_{\alpha,k}(X) \mid \alpha \in K\}$ over k.

Furthermore if K/k is finite then so is L/k

Proof. Suppose $K = k\left(\{\alpha_i\}_{i\in I}\right)$. Let L/k be the splitting field for $S = \left\{m_{\alpha_i,k}(X)\right\}_{i\in I}$. By (3.19.68) L/k is normal and by (3.19.50) there is a morphism $\sigma : K/k \to L/k$, so we may consider it as an extension $(L/K, \sigma)$. Suppose $i_{KL}(K) \subset L' \subset L$ is normal. As $i_{KL}(\alpha_i) \in L'$, by definition S splits in L' and therefore $L' = L$ by (3.19.54). Therefore L/K is a normal closure as required.

If K/k is finite then we may choose I to be finite, and therefore L/k is finite.

Let L/K be an arbitrary normal closure, then we claim L/k is a splitting field for $S' := \{m_{\alpha,k}(X) \mid \alpha \in K\}$. Clearly S' splits in L/k, because each has a root in L. Let L'/k be the subfield generated by roots of S'. Then it is a splitting field and therefore normal by (3.19.68). By assumption $L' = L$ and therefore L/k is the splitting field for S'. Uniqueness follows from the uniqueness of splitting fields (3.19.57). ∎

3.19.8 Separability (Algebraic Case)

We follow Lang and not only characterize separability but define a "separability degree" which equals the extension degree if and only if it's separable. The proofs are somewhat technical, especially in light of the fact most base fields will be perfect.

Definition 3.19.74 (Separable Element)
*We say $\alpha \in K/k$ is **separable** over k if it is algebraic and $m_{\alpha,k}(X)$ is a separable polynomial.*

*We say K/k is **separable algebraic** (or just separable if K/k is assumed to be algebraic) if every $\alpha \in K$ is **separable**.*

Lemma 3.19.75
$\alpha \in K/k$ is separable if and only if it is a root of a separable polynomial in $k[X]$.

In particular α separable over k implies it is separable over any subfield $i_{kK}(k) \subset E \subset K$.

Proof. One direction is obvious. Conversely suppose $f(\alpha) = 0$ with f separable. Then $m_{\alpha,k} \mid f$ so the result follows from (3.17.40). ∎

Proposition 3.19.76 (Separability Degree)
*Let K/k be an algebraic extension and L/K an extension such that L/k is normal (e.g. $L = \overline{k}$ or L is a normal closure). Then define the **separability degree***

$$[K : k]_s := \#\operatorname{Mor}_k(K, L)$$

This is independent of the choice of L/K.

Proof. Given such an L, let L'/K be the intersection of all subfields of L/K normal over k. This is a normal closure of K/k. Let $\sigma \in \operatorname{Mor}_k(K, L)$ and $\alpha \in K$. Then $\sigma(\alpha)$ is a root of $m_{\alpha,k}(X)$ along with $i_{KL}(\alpha) \in L'$. As L' is normal we have $\sigma(\alpha) \in L'$. Therefore without loss of generality we may replace L with L'. As the normal closure is unique up to isomorphism the degree is well-defined. ∎

First we prove a key lemma regarding simple extension

Lemma 3.19.77 (Separability Degree of Simple Extension)
If $k(\alpha)/k$ is a simple extension and L/k normal overfield then

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$$[k(\alpha):k]_s = \#\{\text{roots of } \ m_\alpha \ \text{ in } \ L\} \leq \deg(m_\alpha) = [k(\alpha):k]$$

Furthermore equality holds iff α is separable over k.

Proof. The first equality follows from (3.19.49), the final equality from (3.19.42). The inequality follows from (3.17.35). The final statement follows from (3.19.11). ∎

The main results of this section are the following

Proposition 3.19.78 (Separability Degree)
Let $L/K/F/k$ be a tower of finite extensions with L/k normal. Then the restriction map

$$\mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(F, L)$$

is surjective with finite fibers of order $[K:F]_s$. In particular we have

$$[K:k]_s = [K:F]_s[F:k]_s$$

Further $[K:k]_s \leq [K:k]$ with equality if and only if K/k is separable.

Proof. For a tower $L/K/F/k$ with L/k normal, consider the restriction map

$$\psi := \mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(F, L)$$

It is surjective by (3.19.70). Consider $\sigma \in \mathrm{Mor}_k(F, L)$ and the fibre $\psi^{-1}(\sigma) = \mathrm{Mor}_F(K, (L/F, \sigma))$. This has order equal to $\#\psi^{-1}(\sigma) = [K:F]_s$ for all σ, because by (3.19.76) this does not depend on the embedding i_{FL}. As $\mathrm{Mor}_k(K, L)$ is equal to the disjoint union of all the fibres, then the multiplicativity result follows.

It is possible to decompose K/k as a tower of simple extensions

$$K = K_n/.../K_0 = k$$

with $K_i = K_{i-1}(\alpha_i)$. By (3.19.77) we have

$$[K_i : K_{i-1}]_s \leq [K_i : K_{i-1}]$$

with equality iff α_i separable over K_{i-1}. By multiplicativity the inequality follows.

If K/k is separable then by (3.19.75) α_i is separable over K_{i-1} and we have $[K_i : K_{i-1}]_s = [K_i : K_{i-1}]$ and $[K:k]_s = [K:k]$ by multiplicativity. Conversely if $[K:k]_s = [K:k]$ then $[K_i : K_{i-1}]_s = [K_i : K_{i-1}]$ and α_i is separable over K_{i-1}. Since the choice of α_1 was arbitrary we see that K/k is separable. ∎

Proposition 3.19.79 (Towers of Separable Extensions)
Consider a tower of algebraic extensions $K/E/k$. Then K/E and E/k is separable iff K/k is.

Proof. K/k separable $\implies$ K/E and E/k separable follows from (3.19.75).

Conversely the finite case follows from (3.19.78) by multiplicativity. For the general case, consider $\alpha \in K$. Then (3.19.46) shows the existence of a finite subextension E_0/k of E such that $m_{\alpha,E} = m_{\alpha,E_0}$. Therefore α is separable over E_0. By (3.19.77) we see that $[E_0(\alpha) : E_0]_s = [E_0(\alpha) : E_0]$. As E/k is separable a-fortiori E_0/k is separable so by (3.19.78) $[E_0 : k]_s = [E_0 : k]$. By multiplicativity $[E_0(\alpha) : k]_s = [E_0(\alpha) : k]$ and the same result again shows that $E_0(\alpha)/k$ is separable. In particular α is separable over k as required. ∎

Proposition 3.19.80
An algebraic extension $K = k(\alpha_1, ..., \alpha_n)/k$ is separable iff α_i are.

Proof. Let $K = k(\alpha_1, ..., \alpha_n)$. Then we may construct a tower of finite (simple) extensions

$$K = K_n/.../K_0 = k$$

with $K_i = k(\alpha_1, ..., \alpha_i)$ and $K_i = K_{i-1}(\alpha_i)$. By (3.19.75) α_i is separable over K_{i-1}. Therefore $[K_i : K_{i-1}]_s = [K_i : K_{i-1}]$ by (3.19.77) and $[K:k]_s = [K:k]$ by multiplicativity. (3.19.78) shows that K/k is separable. ∎

Proposition 3.19.81
Let K/k be a splitting field for finitely many separable polynomials f_i. Then K/k is finite and Galois.

Proof. By definition K is generated by the (finitely) many roots α_{ij} and so is finite by (3.19.44) and normal by (3.19.68). By (3.19.75) each α_{ij} is separable and so by (3.19.80) K/k is separable. ∎

Proposition 3.19.82
Let K/k be a finite separable extension and L/k a normal closure. Then L/k is separable.

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Proof. By (3.19.73) L/k is the splitting field of finitely many minimal polynomials $m_{\alpha_i,k}(X)$ for $\alpha_i \in K$. By (3.19.80) the α_i are separable and therefore so are $m_{\alpha_i,k}(X)$. By (3.19.81) we see that L/k is separable. ∎

Proposition 3.19.83 (Equivalent Definition of Separability)
Let K/k be an algebraic extension. TFAE

a) K/k is separable
b) E/k is separable for every finite subextension
c) $[E:k]_s = [E:k]$ for every finite subextension

Proof. $a) \implies b)$ is trivial and $b) \iff c)$ follows from (3.19.78). We need only show $b) \implies a)$.

Consider $\alpha \in K$. Then by (3.19.46) there exists a finite subextension E/k such that α is algebraic over E. Therefore $E(\alpha)/k$ is finite, and by assumption $E(\alpha)/k$ separable as required. ∎

3.19.9 Purely Inseparable Extensions and Separable Closure

In what follows we let p be the characteristic exponent of k. In other words when $\mathrm{char}(k) = 0$ then $p = 1$ and the statements are trivial.

Definition 3.19.84 (Purely Inseparable Element)
*An element $x \in K/k$ is **purely inseparable** (or p-radical) if there exists an integer $n \geq 0$ such that $x^{p^n} \in k$. The **height** of x is the least such integer.*

*We say an extension K/k is **purely inseparable** if every $x \in K/k$ is purely inseparable.*

*Further K/k is **purely inseparable of height $\leq n$** if additionally every $x \in K$ has height at most n.*

Lemma 3.19.85
Let $a \in k$. For every integer $e \geq 0$ the polynomial $f(X) := X^{p^e} - a$ has at most one root in any extension field K/k.

Proof. Suppose that b is a root, then by (3.19.6) we have $f(X) = (X-b)^{p^e}$. Then evidently b is the unique root. ∎

Lemma 3.19.86
Let $a \in k \setminus k^p$. Then for every integer $e \geq 0$ the polynomial $X^{p^e} - a$ is irreducible in $k[X]$.

Proof. Let K/k be a splitting field for $f(X) = X^{p^e} - a$, $b \in K$ a root and $g(X)$ be the minimal polynomial of b. By (3.19.6) $f(X) = (X-b)^{p^e}$ in $K[X]$. Suppose π is a monic irreducible factor of f in $k[X]$ then by unique factorisation in $K[X]$ we have $\pi(X) = (X-b)^r$ for some r. In particular π has b as a root and therefore is divisible by g. By irreducibility we conclude every irreducible factor is equal to g and the irreducible factorization of f is g^s. Furthermore $p^e = rs$. Consequently both r and s are pth powers, and we have

$$\pi = (X-b)^{p^d}$$
$$f = \pi^{p^{e-d}}$$

for some $0 \leq d \leq e$. In particular $b^{p^d} \in k$ and $a = b^{p^{e-d}}$. By assumption a is not a p-th power and so we must have $d = e$, and $f = \pi$ is irreducible. ∎

Proposition 3.19.87
Let $x \in K/k$ be purely inseparable of height e. Then the minimal polynomial of x is $X^{p^e} - x^{p^e}$. Furthermore

$$[k(x):k] = p^e$$
$$[k(x):k]_s = 1$$

More precisely for every field L/k there is at most one morphism $k(x)/k \to L/k$.

Proof. By definition $x^{p^e} \in k$ is not a p-th power. Therefore the given polynomial is irreducible by (3.19.86) and therefore is the minimal polynomial by uniqueness. The first relation follows from (3.19.42) and the second from (3.19.77). ∎

Corollary 3.19.88
Let $x \in K/k$. Then the following are equivalent

a) x is purely inseparable
b) $m_x(X) = (X-x)^r$ for some r

In this case we must have $r = p^n$ where n is the height of x.

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Corollary 3.19.89
Suppose $x \in K/k$ is both separable and purely inseparable. Then $x \in k$.

Proof. The minimal polynomial is $X^{p^e} - x^{p^e} = (X-x)^{p^e}$. Therefore by (3.19.11) $e = 0$ which means precisely $x \in k$. ∎

We first establish the following two results, which will be used below.

Lemma 3.19.90 (Inseparable Polynomials)
Let $f \in k[X]$ be a non-constant polynomial and p the characteristic exponent of k. Then $f' = 0 \iff p > 1$ and $f \in k[X^p]$.

Proposition 3.19.91 (Separable Irreducible Polynomials)
Let $f \in k[X]$ be an irreducible polynomial and p the characteristic exponent of k. Then the following are equivalent

a) f is separable (i.e. f and f' are co-maximal)
b) $f' \neq 0$
c) $p = 1$ or $f \notin k[X^p]$

Proof. Note by assumption f is not a unit and therefore not constant.

$a) \implies b)$ Suppose $f' = 0$ then $(f, f') = (f) \neq k[X]$ and therefore f is not separable.

$b) \implies a)$ Suppose f is not separable. By (...) (f) is maximal and so we must have $f' \in (f)$. As $\deg(f') < \deg(f)$ this implies $f' = 0$.

$b) \iff c)$ This is just the contrapositive of (3.19.90). ∎

Proposition 3.19.92 (Relative Separable Closure)
Let K/k be a field extension. The subextension $K_s/k \subset K/k$ of separable elements form a separable algebraic subfield. Furthermore if K/k is algebraic then K/K_s is purely inseparable and the restriction map

$$\mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(K_s, L)$$

is bijective for every normal extension L/K. In particular when K_s/k is finite then we have the relation

$$[K_s : k] = [K_s : k]_s = [K : k]_s$$

Proof. Suppose $\alpha, \beta \in K$ are separable algebraic then the field $k(\alpha, \beta)$ is separable algebraic by (3.19.80). As this contains $\alpha \pm \beta$ and $\alpha\beta$ then K_s is a subfield which is by definition separable. Then by (3.19.78) we have $[K_s : k] = [K_s : k]_s$ when this is finite.

For $x \in K$ let $f(X)$ be the minimal polynomial over k. There exists some integer $m \geq 0$ such that $f(X) \in k[X^{p^m}]$ but not in $k\left[X^{p^{m+1}}\right]$. In other words $f(X) = g(X^{p^m})$ and $g(X) \notin k[X^p]$. As f is irreducible so is g, and is therefore the minimal polynomial of x^{p^m}. By (3.19.91) g is separable and therefore $x^{p^m} \in K_s$. This shows that K/K_s is purely inseparable.

Let $L/K/k$ be tower such that L/k is normal. Consider the mapping

$$\mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(K_s, L)$$

obtained by restriction. By (3.19.70) it is surjective. Consider $\psi : K_s \to L$ and $\hat{\psi} : K \to L$ an extension. For every $x \in K$ we have $\hat{\psi}|_{K_s(x)} = i_{K_s(x)L}$ by (3.19.87). This shows that $\hat{\psi}$ is unique and the mapping is bijective. ∎

Definition 3.19.93 (Degree of Inseparability)
*Let K/k be a finite extension. Then in light of (3.19.92) we may define the **degree of inseparability** to be the integer*

$$[K : k]_i := \frac{[K : k]}{[K : k]_s}$$

By (3.19.16) and (3.19.78) it is multiplicative in the sense that for a tower of finite extensions

$$[L : k]_i = [L : K]_i [K : k]_i$$

Further by (3.19.78) K/k is separable if and only if $[K : k]_i = 1$.

We may refine this further by showing that $[K : k]_i$ is always a power of the characteristic exponent.

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Proposition 3.19.94
Let $k(\alpha)/k$ be a simple algebraic extension with characteristic exponent p. There exists a unique integer $r \geq 0$ such that every root of $m_{\alpha,k}(X)$ has multiplicity p^r (in any splitting field).

Further

$$[k(\alpha) : k]_i = p^r$$

and $m_{\alpha,k}(X)$ has $[k(\alpha) : k]_s$ distinct roots.

Proof. Let $L/k(\alpha)/k$ be a normal overfield, and let $\alpha_1, ..., \alpha_s$ be the distinct roots of $m_{\alpha,k}(X)$ with $\alpha_1 = \alpha$. Then we have a factorisation in $L[X]$ given by

$$m_{\alpha,k}(X) = \prod_{i=1}^{s} (X - \alpha_i)^{m_i}$$

for some integers $m_i > 0$.

For every $1 \leq i \leq n$ there exists an automorphism $\sigma_i \in \mathrm{Aut}(L/k)$ such that $\sigma_i(\alpha) = \alpha_i$ (3.19.50). By definition σ_i fixes $m_{\alpha,k}(X)$ so by unique factorisation we deduce that $m_i = m_1$, so all the roots have the same multiplicity m.

If $m = 1$ then by (3.17.39) $m_{\alpha,k}(X)$ is separable. Otherwise by (3.19.91) there is a polynomial $g(X)$ such that $f(X) = g(X^p)$ which has α^p as a root. We may continue inductively to find an integer $r \geq 0$ such that $f(X) = g(X^{p^r})$ and $g(X)$ is separable. Then α^{p^r} is separable. By definition α is purely inseparable over $k(\alpha^{p^r})$ so by (3.19.87)

$$\left[k(\alpha) : k(\alpha^{p^r})\right]_i = p^r$$

By (3.19.78) we have

$$\left[k(\alpha^{p^r}) : k\right]_i = 1$$

and by multiplicativity $[k(\alpha) : k]_i = p^r$. By (3.19.77) the number of distinct roots of $m_{\alpha,k}(X)$ is $[k(\alpha) : k]_s$. Comparing degrees we have $\deg\big(m_{\alpha,k}(X)\big) = [k(\alpha) : k] = [k(\alpha) : k]_s \cdot m$ whence $m = [k(\alpha) : k]_i = p^r$ as required. ∎

Proposition 3.19.95
Let K/k be a finite field extension with characteristic exponent p. Then $[K : k]_i = p^r$ for some integer $r \geq 0$.

K/k is separable if and only if $p = 1$ or $r = 0$.

Proof. We may decompose K/k into a tower of simple extensions, and the result follows from multiplicativity and (3.19.94). ∎

3.19.10 Separable closure

Proposition 3.19.96 (Relatively Separably Closed)
Let K/k be an extension. The following are equivalent

a) *Every separable $x \in K/k$ lies in k*
b) *Every separable algebraic subextension is trivial*
c) *K_s/k is trivial*

*In this case we say K/k is **relatively separably closed**.*

If K/k is algebraic then this is equivalent to being purely inseparable.

Proof. The final statement follows from (3.19.92). ∎

Definition 3.19.97
*We say a field k is **separably closed** if every separable algebraic extension is trivial.*

*We say an extension k^{sep}/k is a **separable closure** if it is separable algebraic and separably closed.*

Proposition 3.19.98
Let $\overline{k}/k$ be an algebraic closure, then the subextension $\overline{k}_s/k$ is a separable closure for k.

Furthermore every separable closure is isomorphic to $\overline{k}_s/k$ and $\overline{k}/\overline{k}_s$ is purely inseparable.

Proof. By (3.19.92) $\overline{k}_s/k$ is separable and algebraic. Suppose $K/\overline{k}_s$ is separable and algebraic then by (...) there exists an embedding $i : K/\overline{k}_s \to \overline{k}/\overline{k}_s$. As $\overline{k}_s$ is relatively separably closed in $\overline{k}$ we see that $i(K) = \overline{k}_s$ and therefore $K = \overline{k}_s$.

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Let k^{sep}/k be a separable closure, then by (3.19.61) there is an embedding $\varphi : k^{\text{sep}}/k \to \overline{k}/k$. Then by definition $\varphi(k^{\text{sep}}) \subset \overline{k}_s$, so this defines an extension $\overline{k}_s/k^{\text{sep}}$ which is by definition separable. By assumption it is trivial which means precisely $\varphi(k^{\text{sep}}) = \overline{k}_s$.

It follows from (3.19.92) that $\overline{k}/\overline{k}_s$ is purely inseparable. ∎

Proposition 3.19.99
Let k^{sep}/k be a separable closure and k'/k a separable algebraic extension. Then there exists a morphism

$$k'/k \to k^{\text{sep}}/k$$

Proof. By (3.19.98) we may identify k^{sep}/k with $\overline{k}_s/k$. By (3.19.61) there exists a morphism $k'/k \to \overline{k}/k$ which by definition has image in $\overline{k}_s/k$. ∎

3.19.11 Perfect Fields

Proposition 3.19.100 (Perfect Closure)
Let k be a field with characteristic exponent p contained in an algebraically closed field Ω (e.g. $\overline{k}$). Define

$$k^{p^{-n}} := \left\{x \in \Omega \;\middle|\; x^{p^n} \in k\right\}$$

Then $k^{p^{-n}}$ is a subfield of Ω/k, which is purely inseparable of height $\leq n$. Furthermore it is a splitting field for the family of polynomials $\left\{X^{p^r} - a \;\middle|\; a \in k, r \leq n\right\}$ which characterizes it up to isomorphism. Furthermore

$$k^{p^{-\infty}} := \left\{x \in \Omega \;\middle|\; x^{p^n} \in k \text{ some } n \geq 1\right\} = \bigcup_{n \geq 1} k^{p^{-n}}$$

is the smallest perfect subfield of Ω containing k and is also a splitting field for the family of polynomials $\left\{X^{p^n} - a \;\middle|\; a \in k, n \in \mathbb{N}\right\}$.

Proof. Suppose $\alpha, \beta \in k^{p^{-n}}$ then applying the Frobenius homomorphism (...) we see $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$ and so $k^{p^{-n}}$ is an additive subgroup. Furthermore $(\alpha\beta)^{p^n} = \alpha^{p^n}\beta^{p^n}$ whence it is a multiplicative subgroup as this demonstrates $(\alpha^{-1})^{p^n} = (\alpha^{p^n})^{-1} \in k$. Therefore it is a subfield which is by definition purely inseparable of height $\leq n$.

By definition $k^{p^{-n}}$ is precisely the set of roots of the given polynomials and it is therefore the corresponding splitting field by (...). This also shows that $n \leq m \implies k^{p^{-n}} \subseteq k^{p^{-m}}$ which shows that the union $k^{p^{-\infty}}$ is a subfield of Ω.

To demonstrate it's perfect we need to show every element $x \in k^{p^{-n}}$ has a p-th root. By definition $x^{p^n} \in k$ and there exists a root α to the polynomial $X^{p^{n+1}} - x^{p^n}$ in $k^{p^{-(n+1)}}$. Therefore $(\alpha^p - x)^{p^n} = 0$ whence $\alpha^p = x$ as required.

Let k' be another perfect subfield containing k. We claim by induction that $k^{p^{-n}} \subseteq k'$. For given $\alpha \in k^{p^{-(n+1)}}$ by definition $\alpha^p \in k^{p^{-n}}$ whence $\alpha^p \in k'$ by the inductive hypothesis, and by assumption $\alpha \in k'$. Therefore $k^{p^{-\infty}}$ is the smallest perfect subfield. Furthermore by definition it is the set of roots of the given family of polynomials and therefore the corresponding splitting field. ∎

Proposition 3.19.101
Let k'/k be a purely inseparable extension of height $\leq n$, then there exists an embedding

$$k'/k \to k^{p^{-n}}/k$$

If k'/k is purely inseparable and K/k is perfect then there also exists an embedding

$$k'/k \to K/k$$

Proof. For the first part let $\alpha \in k'$ then by (3.19.87) the minimal polynomial of α is $g(X) = X^{p^e} - \alpha^{p^e}$ for $e \leq n$ which splits completely in $k^{p^{-n}}$ by (3.19.100). Then the embedding exists by (3.19.50).

Suppose K/k is perfect then we claim that $X^{p^n} - a$ splits completely for every $a \in K$. By induction there is $b \in K$ such that $b^{p^n} = a$. Then $X^{p^n} - a = (X - b)^{p^n}$ splits completely. In particular for every $\alpha \in k'$ the minimal polynomial splits completely in K/k and the embedding follows from (3.19.50). ∎

Recall a perfect field k satisfies $k^p = k$ where p is the characteristic exponent. We show that in this case there are no inseparable algebraic extensions. First we show that all finite fields are perfect.

Proposition 3.19.102 (Finite fields are perfect)
Any finite field is perfect.

Proof. The Frobenius homomorphism is injective and therefore surjective by counting. ∎

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Proposition 3.19.103 (Perfect field criteria)
Let k be a field with characteristic exponent p. Then the following are equivalent

a) Every irreducible polynomial in $k[X]$ is separable
b) Every algebraic extension K/k is separable
c) $\overline{k}/k$ is separable
d) $k^p = k$

In particular every field of characteristic zero is perfect, and an algebraic extension of a perfect field is separable.

Proof. We prove each in turn

$a) \implies b)$ Minimal polynomials are irreducible by (3.19.36) and therefore are separable by hypothesis.

$b) \iff c)$ One direction is automatic as $\overline{k}$ is algebraic. On the other hand every algebraic extension is isomorphic to a subfield of $\overline{k}$, so the implication follows from (3.19.79).

$b) \implies d)$ We need only show $k^p = k$ in the case $p > 1$. If there exists $a \in k \setminus k^p$ then by (3.19.86) the polynomial $f(X) := X^p - a$ is irreducible. Then it's clear that the field extension $K := k[X]/(X^p - a)$ is not separable. For the minimal polynomial of $\overline{X}$ is f which by (3.19.91) is not separable. This contradicts the assumption that K/k is separable.

$d) \implies a)$ Suppose that f is irreducible but not separable. Then by (3.19.91) $\operatorname{char}(k) = p > 1$ and $f \in k[X^p]$. By assumption all the coefficients are p-th powers and therefore $f = h^p$ by (3.19.6) for some $h \in k[X]$. This contradicts irreducibility of f, and so f must be separable. ∎

3.19.12 Applications of Separability

Definition 3.19.104 (Bounds on $\operatorname{Aut}(K/k)$)
Let K/k be an algebraic extension and L/K an extension such that L/k is normal. Then there is a natural injection

$$\begin{aligned} \operatorname{Mor}_k(K, i_{KL}) : \operatorname{Aut}(K/k) &\to \operatorname{Mor}_k(K, L) \\ \sigma &\to i_{KL} \circ \sigma \end{aligned}$$

In particular in the case $[K : k] < \infty$

$$\#\operatorname{Aut}(K/k) \leq [K : k]_s \leq [K : k] < \infty$$

If i_{KL} is inclusion, then we may regard $\operatorname{Aut}(K/k)$ as a subset of $\operatorname{Mor}_k(K, L)$

As an application of the concept of separability degree we prove

Proposition 3.19.105 (Primitive Element Theorem)
Let K/k be a finite separable extension of k then $K = k(\alpha)$ is simple.

Proof. We only prove the case k is infinite. The finite case can be proven separately by showing that the $K^\star$ is cyclic.

Consider the set $\operatorname{Mor}_k\left(K, \overline{k}\right) = \{\sigma_1, ..., \sigma_n\}$ which by (3.19.83) has order $n = [K : k]$. By induction we can assume that $K = k(\alpha, \beta)$. We claim that there exists $0 \neq c \in k$ such that $\sigma_i(\alpha + c\beta)$ are all distinct. In this case we clearly have $\#\operatorname{Mor}_k\left(k(\alpha + c\beta), \overline{k}\right) \geq n$ so by the same result $[k(\alpha + c\beta) : k] \geq [k(\alpha + c\beta) : k]_s \geq n$ whence $k(\alpha + c\beta) = K$ (by (2.3.15)).

We have $\sigma_i(\alpha + c\beta) = \sigma_j(\alpha + c\beta) \iff c\big(\sigma_i(\beta) - \sigma_j(\beta)\big) = \big(\sigma_i(\alpha) - \sigma_j(\alpha)\big)$. Therefore consider the polynomial

$$f(X) = \prod_{i \neq j} \Big(X\big(\sigma_i(\beta) - \sigma_j(\beta)\big) - \big(\sigma_i(\alpha) - \sigma_j(\alpha)\big)\Big)$$

Then the embeddings are distinct precisely when $f(c) \neq 0$. Since $f(X)$ has at most finitely many roots and k is infinite, there must exist such a c. ∎

3.19.13 Normal Extensions II

We provide some more straightforward criteria based on $[K : k]_s$

Proposition 3.19.106 (Normal Criteria II)
Let $L/K/k$ be a tower of algebraic extensions such that L/k is normal (e.g. $L = \overline{k}$). Then K/k is normal if and only if the embedding

$$\operatorname{Mor}_k(K, i_{KL}) : \operatorname{Aut}(K/k) \to \operatorname{Mor}_k(K, L)$$

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is a bijection. In particular if K/k is finite, then it is normal if and only if

$$\#\operatorname{Aut}(K/k) = [K:k]_s$$

Proof. Suppose K/k is normal and consider $\sigma \in \operatorname{Mor}_k(K,L)$. Then by **NOR1** $\sigma(K) = i_{KL}(K)$ and we may define $\tau := i_{KL}^{-1} \circ \sigma$ with $\tau \in \operatorname{Aut}(K/k)$. The converse is similar.

For the final part we've already observed (3.19.104) that in the finite case $\#\operatorname{Aut}(K/k) \leq [K:k]_s = \#\operatorname{Mor}_k(K,L) \leq [K:k] < \infty$. Therefore the embedding $\operatorname{Mor}_k(K, i_{KL})$ is a bijection precisely when the orders are the same. ∎

Corollary 3.19.107 (Galois Criteria)
Let K/k be a finite extension. Then

$$\#\operatorname{Aut}(K/k) \leq [K:k]_s \leq [K:k]$$

with equalities if and only if K/k is Galois.

Proof. We've seen the inequalities (3.19.104)

$$\#\operatorname{Aut}(K/k) \leq [K:k]_s \leq [K:k] < \infty$$

with equality if and only if K/k is both normal (3.19.106) and separable (3.19.78) ∎

3.19.14 Finite Fields

A finite field K necessarily has positive characteristic p, and therefore the prime subfield is isomorphic to the field $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$. We list some necessary properties of a finite field

Proposition 3.19.108 (Properties of finite fields)
Every finite field K is a finite-dimensional vector space over its prime subfield $\mathbb{F}_p$. Define $n = \left[K:\mathbb{F}_p\right]$.

- $\#K = p^n$
- *K is a splitting field for $X^{p^n} - X \in \mathbb{F}_p[X]$, and indeed is equal to the set of roots*
- *The multiplicative group of units $K^\star$ is cyclic.*
- *$K/\mathbb{F}_p$ is simple*

Proof. Since $K/\mathbb{F}_p$ is a finite-dimensional vector space it must have order p^n.

The group of units has order $p^n - 1$, so by Lagrange's theorem every non-zero element satisfies $X^{p^n-1} - 1 = 0$, so therefore every element satisfies $X^{p^n} - X = 0$. Since this polynomial can have at most p^n roots (3.17.35) the roots are exactly all the elements of K.

We note again that $X^d - 1$ has at most d roots by (3.17.35). Therefore the fact $K^\star$ is cyclic follows from (3.2.25). ∎

Proposition 3.19.109 (Frobenius morphism)
Given any field $K/\mathbb{F}_p$ the Frobenius map

$$\phi : x \to x^p$$

is an injective field homomorphism. In particular when K is finite (or even algebraic) it is an automorphism over $\mathbb{F}_p$.

Proof. The only non-trivial step is showing

$$(x+y)^p = x^p + y^p$$

which follows from elementary calculations on binomial coefficients.

For the final statement use (3.19.35). ∎

Further we can show existence and uniqueness of finite fields.

Proposition 3.19.110 (Existence and uniqueness of finite fields)
Consider the algebraic closure $\overline{\mathbb{F}_p}$ and let $\mathbb{F}_{p^n}$ denote the splitting field of $f(X) = X^{p^n} - X$ in $\overline{\mathbb{F}_p}$. Then

- *$\mathbb{F}_{p^n}$ is equal to the set of roots of $X^{p^n} - X$*
- *It is the unique subfield of order p^n and every finite field of order p^n is isomorphic to this.*
- *$\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n} \iff m \mid n$*

Proof. By (3.19.109) the set of roots of $f(X)$ forms a subfield of $\overline{\mathbb{F}_p}$.

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Furthermore $f'(X) = -1$ so $f(X)$ is separable because clearly $(f, f') = 1$. Therefore by (3.19.11) $f(X)$ has p^n distinct roots and the splitting field of $f(X)$ is exactly the set of roots.

Furthermore every subfield of order p^n must satisfy this polynomial by Lagrange's Theorem (3.2.12), so it is the unique such subfield.

Since every algebraic extension of $\mathbb{F}_p$ is isomorphic to a subfield of $\overline{\mathbb{F}_p}$ it's also the unique algebraic extension of order p^n up to isomorphism.

Clearly if $\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n}$ then $\left[\mathbb{F}_{p^n} : \mathbb{F}_p\right] = \left[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}\right]\left[\mathbb{F}_{p^m} : \mathbb{F}_p\right]$, so we must have $m \mid n$. Conversely if $\alpha \in \mathbb{F}_{p^m}$ then $\alpha^{p^m} = \alpha \implies \alpha^{p^{rm}} = \alpha$ for all $r > 0$, so $\alpha \in \mathbb{F}_{p^n}$. ∎

It is usually most convenient to work in $\overline{\mathbb{F}_p}$ and consider the subfields of order p^n. We've seen in (3.19.102) that every finite field $\mathbb{F}_q := \mathbb{F}_{p^n}$ is perfect and therefore every algebraic extension is separable (3.19.103). In fact we may show that every finite extension is Galois.

Proposition 3.19.111
The field extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois with

$$\mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right) = \langle \phi \rangle$$

cyclic of order n generated by the Frobenius automorphism.

Proof. Let $G = \mathrm{Aut}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right)$. We've observed that $\phi \in G$. Let $d = o(\phi)$, then we wish to prove that $n = d$. Certainly Lagrange's Theorem applied to the multiplicative group $\mathbb{F}_{p^n}^\star$ implies $\phi^n = 1$. Therefore $d \mid n$ by (3.2.12) applied to G. By definition $\phi^d = e$, so every $\alpha \in \mathbb{F}_{p^n}$ satisfies the polynomial $X^{p^d} - X = 0$. This polynomial has at most p^d roots (3.17.35) so we must have $d \geq n$, and therefore $d = n$. Clearly ϕ generates a cyclic subgroup of order n. However by (3.19.107) G has at most order n, whence $G = \langle \phi \rangle$ as required. Furthermore by the same result $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois. ∎

Proposition 3.19.112 (Subfields of $\mathbb{F}_{p^n}$)
Consider the field extension $\mathbb{F}_{p^n}/\mathbb{F}_p$. Then it has a unique subfield of order p^m if $m \mid n$ (and no such subfield otherwise). In this case $\mathbb{F}_{p^n}/\mathbb{F}_{p^m}$ is Galois and

$$\mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_{p^m}\right) = \langle \phi^m \rangle$$

and in particular has order n/m.

Proof. We've already shown that $\mathbb{F}_{p^n}$ has a unique subfield of order p^m, by assuming an embedding in $\overline{\mathbb{F}_p}$. Let $H = \mathrm{Aut}\left(\mathbb{F}_{p^n}/\mathbb{F}_{p^m}\right)$. Note ϕ^m has order n/m. Furthermore from (3.19.110) every element of $\mathbb{F}_{p^m}$ satisfies $X^{p^m} - X$. In other words ϕ^m fixes $\mathbb{F}_{p^m}$ and $\phi^m \in H$. Therefore $\langle \phi^m \rangle \leq H$ and $\#H \geq n/m$. By (3.19.107) $\#H \leq \left[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}\right] = n/m$, whence we have equality and so the extension is Galois by (3.19.107) and $H = \langle \phi^m \rangle$. ∎

Lemma 3.19.113
Let $x \in \mathbb{F}_{p^n}$. Then $\deg(x) = n \iff \mathrm{Fix}(x) = \{1\} \subset \mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right)$.

In other words $\mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right)$ *acts freely on the elements of degree n.*

Proof. Observe that $\deg(x) = n \iff \left[\mathbb{F}_p(x) : \mathbb{F}_p\right] = n \iff \mathbb{F}_p(x) = \mathbb{F}_{p^n} \overset{(3.19.115)}{\iff} \mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_p(x)\right) = \{1\}$ which is equivalent to the statement given. ∎

3.19.15 Galois Theory

We've seen that for K/k a finite extension

$$\#\,\mathrm{Aut}(K/k) \leq [K : k]_s \leq [K : k]$$

with equality if and only if K/k is Galois, by (3.19.107).

Remark 3.19.114
If k is perfect then $\overline{k}/k$ is Galois.

The main result of Galois Theory is

Proposition 3.19.115
Let K/k be a finite Galois extension then there is an order-reversing bijection between subgroups and subfields

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$$\begin{aligned}\{H \leq \mathrm{Gal}(K/k)\} &\longleftrightarrow \{F/k \subseteq K/k\} \\ H &\longrightarrow K^H := \{x \in K \mid h(x) = x \quad \forall h \in H\} \\ \mathrm{Gal}(K/F) &\longleftarrow F\end{aligned}$$

Proof. This is proved in a series of Propositions in the rest of this section. Firstly we show it is well-defined in (3.19.116). The maps are mutual inverses by (3.19.117) and (3.19.118). ∎

Such an order reversing map is usually called an (antitone) Galois connection, as the first such type arose from Galois Theory. Note it is well-defined because of the following proposition.

Proposition 3.19.116
If K/k is Galois and $F \subset K/k$ then K/F is Galois.

Proof. This follows from (3.19.65) and (3.19.79). ∎

Proposition 3.19.117 (Fixed field of Galois group)
If K/k is Galois and $F \subset K/k$ a subfield then

$$K^{\mathrm{Gal}(K/F)} = F$$

Proof. Clearly $F \subseteq K^{\mathrm{Gal}(K/F)}$. Conversely given $\alpha \in K \setminus F$, then $\deg(m_{\alpha,F}) > 1$. Since α is separable it must have another root $\beta \in K$. By (3.19.66) there is an element $\sigma \in \mathrm{Gal}(K/F)$ such that $\sigma(\alpha) = \beta$. In other words $\alpha \notin K^{\mathrm{Gal}(K/F)}$, which shows the reverse inclusion. ∎

Proposition 3.19.118
Let K/k be a field extension and $H \subseteq \mathrm{Aut}(K/k)$ a finite subgroup then K/K^H is a finite Galois extension with

$$\mathrm{Gal}(K/K^H) = H$$

and order equal to $[K : K^H]$. When K/k is finite then H is automatically finite.

Proof. Firstly observe that trivially $H \subseteq \mathrm{Aut}(K/K^H)$. If we know that $[K : K^H] < \infty$, then by (3.19.107) we have

$$\#H \leq \#\,\mathrm{Aut}(K/K^H) \leq [K : K^H]_s \leq [K : K^H]$$

We can prove equality everywhere if we show that $[K : K^H] \leq \#H$, which is shown either by (3.19.119) or (3.19.120). Note equality also shows that K/K^H is finite Galois by the same result.

Finally when K/k is finite, then $\#\,\mathrm{Aut}(K/k) < \infty$. So in this case H is always finite. ∎

We present two approaches to showing the inequality $[K : K^H] \leq \#H$. The first uses independence of characters style argument, and the second which is more straightforward uses the action of H to show that every element has degree at most $\#H$ (Artin).

Lemma 3.19.119 (Bound degree of fixed field I)
Let K/k be a field extension and $H \subset \mathrm{Aut}(K/k)$ a finite subgroup. Then

$$[K : K^H] \leq \#H$$

Proof. Let $H = \{\sigma_1, ..., \sigma_n\}$ with $\sigma_1 = \mathrm{id}$ and $\{\alpha_1, ..., \alpha_m\}$ a K^H-linearly independent subset of K.

Consider the vector space K^n and the elements $\hat{\alpha}_j = \left(\sigma_1(\alpha_j), ..., \sigma_{n(\alpha_j)}\right)$ for $j = 1...m$. It's enough to show that these are linearly independent over K, as that implies $m \leq n$ by (3.3.126). Therefore K/K^H is finite dimensional with dimension at most $n = \#H$.

Let $S(K) := \left\{v \in K^m \;\middle|\; \sum_{j=1}^m v_j\hat{\alpha}_j = 0\right\}$, we aim to show that $S(K) = \{0\}$. If we also consider $S(K^H)$, any non-zero elements will be a K^H linear-dependence for $\alpha_1, ..., \alpha_m$ by considering the first component ($\sigma_1 = \mathrm{id}$). Therefore by linear independence of α_i we see $S(K^H) = \{0\}$. So it's enough to show that $S(K) \neq \{0\} \implies S(K^H) \neq \{0\}$, to prove $S(K) = \{0\}$ by contradiction.

First observe that $K^\star$ and H both act on $S(K)$ component-wise. The first by multiplication and the second by application. This is well-defined because $v \in S(K)$ if and only if

$$\sum_j v_j\sigma(\alpha_j) = 0 \quad \forall \sigma \in H.$$

Apply τ to obtain

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$$\sum_j \tau(v_j)(\tau\circ\sigma)(\alpha_j) = 0 \quad \forall \sigma \in H$$

and since multiplication by τ permutes H we see $\tau(v) \in S(K)$ as required.

If there exists $0 \neq v \in S(K)$, consider v with a minimal number of non-zero components. By scaling we can assume λv has at least one component in K^H. The vector $\tau(\lambda v) - \lambda v$ then has at least one fewer non-zero components, so by minimality must be zero. Since τ was arbitrary we see $0 \neq \lambda v \in S(K^H)$ as required. ∎

Lemma 3.19.120 (Bound degree of fixed field II)
Let K/k be a field extension and H a finite subgroup of $\mathrm{Aut}(K/k)$. Then K/K^H is finite separable, and simple, with $[K : K^H] \leq \#H$

Proof. We show that K/K^H is separable and every element has degree at most $\#H$. For any $\alpha \in K$, consider the orbit $\alpha^H = \{\sigma(\alpha) \mid \sigma \in H\}$, which is of order at most $\#H$. Then the polynomial

$$f(X) = \prod_{\beta \in \alpha^H} (X - \beta)$$

has α as a root and is separable by (3.19.11). Furthermore $f^\tau = f$ because τ permutes α^H (it's injective and hence bijective). Therefore $f \in K^H[X]$ and $m_{\alpha,K^H} \mid f$. We see that α has degree at most $\#H$ and is separable by (3.17.40).

If K/k is finite, then a-fortiori K/K^H is finite, so we may apply the Primitive Element Theorem (3.19.105) directly to show the result.

More generally let $K^H(\alpha)$ be a simple subfield of K of maximal degree. This exists because the degree of α is bounded above by $\#H$. We claim $K^H(\alpha) = K$, for if not then $K^H \subseteq K^H(\alpha) \subsetneq K^H(\alpha, \beta)$ is a finite separable extension of K^H, whence it must be simple by the Primitive Element Theorem (3.19.105), contradicting maximality. Finally the degree of $[K : K^H]$ is the degree of α, which we've seen is bounded above by $\#H$. ∎

Now we may demonstrate straightforward criteria for subfield to be normal

Proposition 3.19.121
Let K/k be a finite Galois extension and $k \subset F \subset K$ a subfield.

Then F/k is Galois if and only if $\mathrm{Gal}(K/F) \lhd \mathrm{Gal}(K/k)$ is normal. In this case we have a canonical isomorphism

$$\mathrm{Gal}(K/k)/\mathrm{Gal}(K/F) \to \mathrm{Gal}(F/k)$$

Proof. Recall from (3.19.67) we have F/k is normal iff $\sigma(F) = F$ for all $\sigma \in \mathrm{Gal}(K/k)$. Recall K/F is normal for all subfields F. Furthermore, we observe that

$$\mathrm{Gal}(K/\sigma(F)) = \sigma\,\mathrm{Gal}(K/F)\sigma^{-1}$$

By the correspondence (3.19.115) $\mathrm{Gal}(K/F) = \mathrm{Gal}(K/F') \iff F = F'$. Therefore

$$\begin{aligned} F/k \text{ normal} &\iff \sigma(F) = F \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \mathrm{Gal}(K/\sigma(F)) = \mathrm{Gal}(K/F) \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \sigma\,\mathrm{Gal}(K/F)\sigma^{-1} = \mathrm{Gal}(K/F) \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \mathrm{Gal}(K/F) \lhd \mathrm{Gal}(K/k) \end{aligned}$$

The result then follows from (3.19.71). ∎

3.19.16 Norm and Trace

In what follows we assume that K is a commutative ring and A is a K-algebra which is finite free as a K-module (free is automatic when K is a field).

Definition 3.19.122 (Norm, Trace and Characteristic Polynomial)
For $\alpha \in A$ consider the K-linear multiplication map

$$\mathrm{mult}_{\alpha,K} : A \to A$$

and define

a) $\mathrm{Tr}_{A/K}(\alpha) := \mathrm{Tr}(\mathrm{mult}_\alpha) \in K$
b) $\mathrm{Norm}_{A/K}(\alpha) := \det(\mathrm{mult}_\alpha) \in K$
c) $\mathrm{Pc}_{A/K}(\alpha; X) := \chi_{\mathrm{mult}_\alpha} = \det\left(\mathrm{mult}_{X-\alpha,K[X]}\right) = \det(X \cdot I_n - [\mathrm{mult}_\alpha]) \in K[X]$ *the characteristic polynomial.*

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Proposition 3.19.123
With the notation as in (3.19.122) we have the following properties

a) $\mathrm{Tr}(\alpha+\beta) = \mathrm{Tr}(\alpha) + \mathrm{Tr}(\beta)$
b) $\mathrm{Tr}(\alpha\beta) = \mathrm{Tr}(\beta\alpha)$
c) $\mathrm{Norm}(\alpha\beta) = \mathrm{Norm}(\alpha)\,\mathrm{Norm}(\beta)$

and further that

$$\mathrm{Pc}(\alpha; X) = X^n - \mathrm{Tr}(\alpha)X^{n-1} + \cdots + (-1)^n\,\mathrm{Norm}(\alpha)$$

Finally for $\lambda \in K$ *we have*

$$\mathrm{Tr}(\lambda) = \dim_K A \cdot \lambda$$
$$\mathrm{Norm}(\lambda) = \lambda^{\dim_K A}$$

Proof. We prove each in turn

a) We have $\mathrm{mult}_{\alpha+\beta} = \mathrm{mult}_\alpha + \mathrm{mult}_\beta$ and ordinary trace is additive (3.5.24).
b) When A is commutative this is obvious, however even in general we have $\mathrm{mult}_{\alpha\beta} = \mathrm{mult}_\alpha \circ \mathrm{mult}_\beta$ and so the result follows from symmetry of the ordinary trace (3.5.25)
c) This follows similarly because the determinant is multiplicative (3.5.17).

The identity for the characteristic polynomial follows directly from (3.18.7). Finally the matrix representation of mult_λ is simply $\lambda \cdot I_n$ with respect to any basis. ∎

Proposition 3.19.124 (Characteristic Polynomial is a Norm)
For $\alpha \in A$ *we have the relationship*

$$\mathrm{Pc}_{A/K}(\alpha; X) = \mathrm{Norm}_{A[X]/K[X]}(X - \alpha)$$

Proof. This follows from the definitions and (3.18.7). ∎

Proposition 3.19.125
Let M *be a finite free* A*-module. Then* M *is a finite free* K*-module. Explicitly given a basis* (m_j) *of* M *over* A *and* (a_i) *of* A *over* K*, then* $(a_i \cdot m_j)$ *is a basis for* M *over* K*.*

If $u : M \to M$ *is an endomorphism then we denote by* u_K *the endomorphism of* M *regarded as a (finite free)* K*-module.*

Proposition 3.19.126 (Transitivity of Trace)
Suppose A *is commutative,* M *is a finite free* A*-module and* $u : M \to M$ *an endomorphism. Then*

$$\mathrm{Tr}(u_K) = \mathrm{Tr}_{A/K}(\mathrm{Tr}(u))$$

and

$$\det(u_K) = \mathrm{Norm}_{A/K}(\det(u))$$

Proof. Let $(e_1, ..., e_m)$ be an A-basis for M and $(a_1, ..., a_n)$ be a K-basis for A. Then by (3.19.125) $(a_i \cdot e_j)$ is a K-basis for M. For $\alpha \in A$ let $M(\alpha) \in \mathrm{Mat}_n(K)$ denote the matrix corresponding mult_α and let $(c_{lj}) \in \mathrm{Mat}_m(A)$ be the matrix of u with respect to (e_j). By definition this means

$$u(e_j) = \sum_{l=1}^{m} c_{lj} \cdot e_l$$

and

$$a_i\alpha = \alpha a_i = \sum_{k=1}^{n} M(\alpha)_{ki} a_k.$$

Therefore

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$$\begin{aligned} u_K(a_i e_j) &= a_i u(e_j) \\ &= a_i \cdot \left(\sum_{l=1}^{m} c_{lj} \cdot e_l \right) \\ &= \sum_{l=1}^{m} (a_i c_{lj}) \cdot e_l \\ &= \sum_{l=1}^{m} \left(\sum_{k=1}^{n} M(c_{lj})_{ki} a_k \right) \cdot e_l \\ &= \sum_{l=1}^{m} \sum_{k=1}^{n} M(c_{lj})_{ki} \cdot (a_k \cdot e_l) \end{aligned}$$

Consequentally the matrix for u_K with respect to the basis $(a_i \cdot e_j)$ is given in block form by

$$\begin{pmatrix} M(c_{11}) & \dots & M(c_{1m}) \\ \vdots & & \vdots \\ M(c_{m1}) & \dots & M(c_{mm}) \end{pmatrix}$$

Therefore we see that

$$\begin{aligned} \mathrm{Tr}(u_K) &= \sum_{j=1}^{m} \mathrm{Tr}(M(c_{jj})) \\ &= \sum_{j=1}^{m} \mathrm{Tr}_{A/K}(c_{jj}) \\ &= \mathrm{Tr}_{A/K} \left(\sum_{j=1}^{m} c_{jj} \right) \\ &= \mathrm{Tr}_{A/K}(\mathrm{Tr}(u)) \end{aligned}$$

as required. Similarly by (3.5.33)

$$\det(u_K) = \det(D_m(M(c_{11}), \dots, M(c_{mm})))$$

and by the Laplace Expansion

$$\begin{aligned} D_m(M(c_{11}), \dots, M(c_{mm})) &= \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{i=1}^{m} M\left(c_{i\sigma(i)}\right) \\ &= \sum_{\sigma \in S_m} \varepsilon(\sigma) M \left(\prod_{i=1}^{m} c_{i\sigma(i)} \right) \\ &= M \left(\sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{i=1}^{m} c_{i\sigma(i)} \right) \\ &= M(\det(u)) \end{aligned}$$

where we implicitly use that $\alpha \to \mathrm{mult}_\alpha \to M(\alpha)$ is a ring homomorphism. Therefore by definition

$$\det(u_K) = \det(M(\det(u))) = \mathrm{Norm}_{A/K}(\det(u))$$

■

Corollary 3.19.127 (Transitivity)
Suppose B is a commutative A-algebra which is finite free as an A-module. Then

$$\mathrm{Tr}_{B/K}(b) = \mathrm{Tr}_{A/K}\left(\mathrm{Tr}_{B/A}(b)\right)$$

and

$$\mathrm{Norm}_{B/K}(b) = \mathrm{Norm}_{A/K}\left(\mathrm{Norm}_{B/A}(b)\right)$$

for all $b \in B$. Furthermore

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$$\mathrm{Pc}_{B/K}(b;X) = \mathrm{Norm}_{A[X]/K[X]}\big(\mathrm{Pc}_{B/A}(b;X)\big)$$

Proposition 3.19.128
Let K/k be a finite field extension and $\alpha \in K$. Then

$$\mathrm{Pc}_{K/k}(\alpha;X) = m_{\alpha,k}(X)^{[K:k(\alpha)]}$$

Proof. Consider first the case $K = k(\alpha)$. Then by (3.18.4) we have $\mathrm{Pc}_{K/k}(\alpha;X)(\mathrm{mult}_\alpha) = 0$ and in particular $\mathrm{Pc}_{K/k}(\alpha;X)(\alpha) = 0$ (since $\alpha^n = \alpha^n \cdot 1$). By (...) $m_{\alpha,k}$ divides $\mathrm{Pc}_{K/k}(\alpha;X)$. Further by (...) $\deg\big(m_{\alpha,k}\big) = [k(\alpha):k] = [K:k] = \deg\big(\mathrm{Pc}_{K/k}(\alpha;X)\big)$. Since they are both monic then we find they are equal.

For the general case we may write the matrix for mult_α in block form by considering the tower basis for $K/k(\alpha)/k$ (see (3.19.125)). This is block diagonal, so we may use the Leibniz formula to compute this directly as product of determinants of the blocks. ∎

We may pull together these results to give a more classical expression for norm and trace.

Proposition 3.19.129
Let $L/K/k$ be a tower of extensions such that L/k is normal (e.g. $L = \overline{k}$). Then for any $\alpha \in K$ we have the identities

$$i_{kL}\big(\mathrm{Norm}_{K/k}(\alpha)\big) = \left(\prod_{\sigma\in \mathrm{Mor}_k(K,L)} \sigma(\alpha)\right)^{[K:k]_i}$$

and

$$i_{kL}\big(\mathrm{Tr}_{K/k}(\alpha)\big) = [K:k]_i \sum_{\sigma\in \mathrm{Mor}_k(K,L)} \sigma(\alpha)$$

Proof. Recall from (3.19.78) that the restriction map

$$\mathrm{Mor}_k(K,L) \to \mathrm{Mor}_k(k(\alpha),L)$$

is surjective with finite fibers of order $[K:k(\alpha)]_s$. Therefore we may write the right hand side as

$$\begin{aligned}\left(\prod_{\sigma\in \mathrm{Mor}_k(K,L)} \sigma(\alpha)\right)^{[K:k]_i} &= \left(\prod_{\sigma\in \mathrm{Mor}_k(k(\alpha),L)} \sigma(\alpha)\right)^{[K:k(\alpha)]_s[K:k]_i} \\ &= \left(\prod_{\sigma\in \mathrm{Mor}_k(k(\alpha),L)} \sigma(\alpha)\right)^{[K:k(\alpha)][k(\alpha):k]_i}\end{aligned}$$

Similarly by (3.19.127) and (3.19.123) the left hand side may be written as

$$\begin{aligned}\mathrm{Norm}_{K/k}(\alpha) &= \mathrm{Norm}_{k(\alpha)/k}\big(\mathrm{Norm}_{K/k(\alpha)}(\alpha)\big) \\ &= \mathrm{Norm}_{k(\alpha)/k}\big(\alpha^{[K:k(\alpha)]}\big) \\ &= \mathrm{Norm}_{k(\alpha)/k}\,(\alpha)^{[K:k(\alpha)]}\end{aligned}$$

Comparing these expressions shows we may reduce to the case $K = k(\alpha)$.

From (3.19.49) and (3.19.94) the right hand side is the product of all the roots of $m_{\alpha,k}$ including multiplicity, which is $(-1)^{[K:k]} i_{kL}(c_0)$ where c_0 is the constant term of $m_{\alpha,k}(X)$. Then using (3.19.128) and (3.19.123) we see that this also equals $i_{kL}\big(\mathrm{Norm}_{k(\alpha)/k}(\alpha)\big)$ as required.

The case of trace is similar. ∎

Lemma 3.19.130 (Dedekind's Lemma)
Let K/k be a field extension and A a k-algebra. The set of k-algebra homomorphisms

$$\mathrm{AlgHom}_k(A,K)$$

is K-linearly independent as a subset of $\mathrm{Hom}_k(A,K)$.

Proof. Suppose that the k-algebra homomorphisms are not linearly independent. Then choose a minimal subset of distinct k-algebra homomorphisms $\varphi_1, ..., \varphi_n$ with a non-trivial relationship over K

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$$\sum_{i=1}^{n} \lambda_i \varphi_i = 0 \quad \lambda_i \in K^\star$$

If $n = 1$ then $\varphi_1 = 0$ which is a contradiction. Suppose without loss of generality that $n > 1$. For a given $x \in K$ we may define

$$\hat{\varphi}_i := (\varphi_i(x) - \varphi_n(x))\varphi_i \quad i = 1...n-1$$

We may choose x such that at least one is non-zero. Then for all $y \in A$ we have

$$\sum_{i=1}^{n-1} \lambda_i \hat{\varphi}_i(y) = \sum_{i=1}^{n} \lambda_i (\varphi_i(x) - \varphi_n(x))\varphi_i(y) = \sum_{i=1}^{n} \lambda_i \varphi_i(xy) - \varphi_n(x) \sum_{i=1}^{n} \lambda_i \varphi_i(y) = 0$$

This contradicts minimality of the subset. ■

Proposition 3.19.131
Let K/k be a finite extension. Then the following are equivalent

a) K/k is separable
b) $\mathrm{Tr}_{K/k}$ is not identically zero
c) The bilinear form

$$(x, y) \to \mathrm{Tr}_{K/k}(xy)$$

is a perfect pairing (see (3.3.148)).

Proof. c) $\implies$ b) By (3.3.148) for all $0 \neq x \in K$ we have $y \to \mathrm{Tr}_{K/k}(xy)$ is non-zero, so there exists y such that $\mathrm{Tr}_{K/k}(xy) \neq 0$ as required.

b) $\implies$ c) By (3.3.149) it's enough to show that for all $0 \neq x \in K$ the map $y \to \mathrm{Tr}_{K/k}(xy)$ is non-zero. By assumption there exists some z such that $\mathrm{Tr}_{K/k}(z) \neq 0$ then we may consider $y = z/x$.

a) $\implies$ b) By (3.19.93) $[K : k]_i = 1$, and so from (3.19.129) the trace equals $\sum_\sigma \sigma$, which is non-zero by (3.19.130).

b) $\implies$ a) Suppose K/k is not separable then $[K : k]_i$ is a power of p (3.19.95), and so the trace is zero by (3.19.129). ■

3.19.17 Transcendental Field Extensions

Definition 3.19.132 (Algebraic Independence)
*Let K/k be a field extension and $S \subset K$. We say S is **algebraically independent** over k if for every finite subset of distinct elements $x_1, ..., x_n \in S$ we have*

$$f(x_1, ..., x_n) = 0 \implies f = 0$$

for all $f \in k[X_1, ..., X_n]$.

For a subset $S \subset K$ define the closure operator

$$c(S) := \overline{k(S)} \cap K := \{x \in K \mid x \text{ algebraic over } k(S)\}$$

*We say Γ is **algebraically spanning** if $c(\Gamma) = K$, equivalently if $K/k(\Gamma)$ is algebraic.*

*We say that $\mathcal{B}$ is a **transcendence base** if it is both algebraically independent and spanning (i.e. $K/k(\mathcal{B})$ is algebraic).*

Essentially we show that (K, c) satisfies the properties of a matroid, in analogy with vector spaces, so that we can use the results of the Matroids section to show that transcendence bases exist and they satisfy certain properties.

Proposition 3.19.133
Let K/k be a field extension and S, T subsets of K. Then the following are equivalent

a) $S \cup T$ is algebraically independent and $S \cap T = \emptyset$
b) S is algebraically independent over k and T is algebraically independent over $k(S)$
c) T is algebraically independent over k and S is algebraically independent over $k(T)$

Proof. By symmetry it's enough to show that a) $\iff$ b).

Suppose a) holds, then a-fortiori S is algebraically independent over k. Suppose T is algebraically dependent over $k(S)$, then there exists $t_1, ..., t_n \in T$ such that $F(t_1, ..., t_n) = 0$ for some $0 \neq F \in k(S)[X_1, ..., X_n]$. By clearing denominators we may assume that $F \in k[S][X_1, ..., X_n]$. As there are only finitely many coefficients we may also take $S = \{s_1, ..., s_m\}$ to be finite. Explicitly

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$$F(X_1, ..., X_n) = \sum_{\alpha \in \mathbb{N}^n} \lambda_\alpha X_1^{\alpha_1} ... X_n^{\alpha_n}$$

and

$$\lambda_\alpha = G_\alpha(s_1, ..., s_m)$$

for some $G_\alpha \in k[Y_1, ..., Y_m]$. We may then define $\hat{F} \in k[X_1, ..., X_n, Y_1, ..., Y_m]$ by

$$\hat{F}(X_1, ..., X_n, Y_1, ..., Y_m) := \sum_{\alpha \in \mathbb{N}^n} G_\alpha(Y_1, ..., Y_m) X_1^{\alpha_1} ... X_n^{\alpha_n}$$

Then $\hat{F}(t_1, ..., t_n, s_1, ..., s_m) = 0$, and by assumption this is an algebraic dependence (as $S \cap T = \emptyset$). Therefore by assumption $\hat{F} = 0$ and in particular $G_\alpha(s_1, ..., s_m) = 0$ for all α. This implies $F = 0$, a contradiction.

b) $\implies$ *a)* Evidently $S \cap T = \emptyset$ otherwise there is a trivial algebraic dependence of T on $k(S)$. Suppose there is an algebraic dependence of $S \cup T$ on k. We may assume wlog that $S = \{s_1, ..., s_m\}$ and $T = \{t_1, ..., t_n\}$ are finite. Then the evaluation homomorphism

$$k[X_1, ..., X_n, Y_1, ..., Y_m] \to K$$

may be written as a composite

$$k[X_1, ..., X_n, Y_1, ..., Y_m] \to k(S)[X_1, ..., X_n] \to K$$

which are both injective by hypothesis. ∎

Corollary 3.19.134 (Exchange Property)
Let K/k be a field extension, $\Gamma \subseteq K$. Suppose x is algebraic over $k(\Gamma \cup \{y\})$ and transcendental over $k(\Gamma)$, then y is algebraic over $k(\Gamma \cup \{x\})$.

Proof. By considering the extension $K/k(\Gamma)$ we may reduce to the case $\Gamma = \emptyset$.

It's enough to show that x transcendental over k and y transcendental over $k(x)$ implies x transcendental over $k(y)$. This follows directly from (3.19.133) by considering $S = \{x\}$ and $T = \{y\}$. ∎

Corollary 3.19.135 (Extension Property)
Let K/k be a field extension, $S \subseteq K$ algebraically independent and $x \in K$ transcendental over $k(S)$. Then $S \cup \{x\}$ is algebraically independent.

Proof. Follows immediately from (3.19.133). ∎

Lemma 3.19.136 (Embeddings Preserve Transcendence Degree)
Let K/k, L/k be field extensions of finite transcendence degree and $\phi : K \to L$ a k-algebra homomorphism. Then $\operatorname{trdeg}(K/k) \leq \operatorname{trdeg}(L/k)$.

Proof. As ϕ is a ring homomorphism from a field, $\ker(\phi)$ is a proper ideal of K, hence 0, so ϕ is injective. Let $S \subset K$ be a transcendence base of K/k, so $|S| = \operatorname{trdeg}(K/k)$ (3.19.141); as ϕ is injective, $|\phi(S)| = |S|$. Given distinct $s_1, ..., s_n \in S$ and $f \in k[X_1, ..., X_n]$ with $f(\phi(s_1), ..., \phi(s_n)) = 0$: as ϕ fixes k, $\phi(f(s_1, ..., s_n)) = f(\phi(s_1), ..., \phi(s_n)) = 0$, so $f(s_1, ..., s_n) = 0$ by injectivity of ϕ, whence $f = 0$ by algebraic independence of S over k; so $\phi(S)$ is algebraically independent over k. By (3.19.140) applied to L/k, $\operatorname{trdeg}(K/k) = |S| = |\phi(S)| \leq \operatorname{trdeg}(L/k)$. ∎

Corollary 3.19.137 (Equivalent form of independence)
Let K/k be a field extension and $S \subset K$. Then the following are equivalent

a) S is algebraically independent
b) x is transcendental over $k(S \setminus \{x\})$ for all $x \in S$

In other words the algebraically independent subsets are precisely the matroid independent subsets.

Proof. a) $\implies$ *b)* Follows from (3.19.133).

b) $\implies$ *a)* Let $\mathcal{F}$ be the family of algebraically independent subsets. We've shown that all such sets are also matroid independent, and by definition the family is of finite character. Furthermore by (3.19.135) it satisfies the extension property. Therefore by (2.3.6) it is precisely the family of matroid independent sets. ∎

Proposition 3.19.138 (Transcendence Base Exists)
Let K/k be a field extension. Suppose S is an algebraic independent subset of K and $\Gamma \supset S$ is such that $K/k(\Gamma)$ is algebraic. Then there exists a transcendence base $\mathcal{B}$ such that $S \subseteq \mathcal{B} \subseteq \Gamma$.

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Proof. This is (2.3.7). ■

Proposition 3.19.139 (Transcendence Base)
Let K/k be a field extension and $S \subset K$ a subset. Then the following are equivalent

*a) S is a **transcendence base***
b) S is a maximal algebraically independent set
c) S is minimal under the condition $K/k(S)$ is algebraic

*When K/k admits a finite algebraic spanning set then all bases are finite of the same size (**transcendence degree**). Write this as $\text{trdeg}(K/k)$ or $\text{trdeg}_k(K)$.*

Proof. Follows from (2.3.8) and (2.3.12). ■

Proposition 3.19.140
Let K/k be a field extension of finite transcendence degree and $S \subset K$ a subset. Then

- *S algebraically independent $\implies |S| \leq \text{trdeg}(K/k)$*
- *$K/k(S)$ is algebraic $\implies |S| \geq \text{trdeg}(K/k)$*

Proof. Follows directly from (2.3.13). ■

Proposition 3.19.141
Let K/k be a field extension of finite transcendence degree and $S \subset K$ a subset. Then the following are equivalent

- *S is a transcendence base*
- *S is algebraically independent and $|S| \geq \text{trdeg}(K/k)$*
- *$K/k(S)$ is algebraic and $|S| \leq \text{trdeg}(K/k)$*

In this case $|S| = \text{trdeg}(K/k)$.

Proof. Follows directly from (2.3.14). ■

In case K/k is finitely generated then we guarantee that $K/k(S)$ is finite.

Proposition 3.19.142
Let K/k be a finitely generated field extension. Then

- $\text{trdeg}(K/k) < \infty$
- *Suppose $K/k(\Gamma)$ is algebraic then it is in fact finite.*

Proof. a) If K/k is finitely generated then the set of generators is algebraically spanning and so $\text{trdeg}(K/k) < \infty$ by (3.19.139)

b) A-fortiori $K/k(\Gamma)$ is finitely generated and by assumption algebraic so by (...) it is finite ■

3.19.18 Separating Transcendence Base

In applications the existence of a separating transcendence base is important, and is guaranteed when k is perfect, or a weaker condition.

Definition 3.19.143 (Separating Transcendence Base)
*A field extension K/k is **separably generated** if there exists a transcendence base S such that $K/k(S)$ is separable (and algebraic). We call such an S a **separating transcendence base**.*

Note when $\text{char}(k) = 0$ then every transcendence base is separating.

Definition 3.19.144 (Maclane's Criterion)
*We say that an algebra A/k of characteristic exponent p satisfies **Maclane's Criterion** if it is reduced and for all $a_1, ..., a_n \in A$ the following property holds*

$$\{a_1, ..., a_n\} \text{ k-linearly independent} \implies \{a_1^p, ..., a_n^p\} \text{ k-linearly independent}$$

When $p = 1$ this is trivially always satisfied and when $p > 1$ such an algebra is automatically reduced by the following result

Lemma 3.19.145
Let A be a k-algebra and $m > 1$ an integer. Then the following are equivalent

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a) A is reduced
b) $x^m = 0 \implies x = 0$

In particular if A has characteristic exponent $p > 1$ then it is reduced iff the Frobenius homomorphism is injective.

Proof. $a) \implies b)$ is trivial. Conversely suppose $x^n = 0$. Then there exists $a > 0$ such that $n \leq am$. In particular $x^{am} = 0$ and therefore by induction $x = 0$ as required. ∎

This is a generalization of the case of a perfect base field by the following result.

Proposition 3.19.146 (Perfect $\implies$ Maclane)
Let A/k be a reduced algebra with k perfect. Then it satisfies Maclane's Criterion.

In particular this holds for every field extension K/k of a perfect base field.

Proof. By assumption A is reduced so we may consider only the case $p > 1$.

Suppose $0 = \sum_i \lambda_i a_i^p$ then by hypothesis $\lambda_i = \mu_i^p$. By (3.19.6) $0 = \left(\sum_i \mu_i a_i\right)^p$. As A is reduced we find $0 = \sum_i \mu_i a_i$ as required. ∎

Lemma 3.19.147
Let $n \geq 0$ an integer and $K = k(x_1, ..., x_{n+1})$ an extension of k such that

- $\text{char}(k) \neq 0$
- $\{x_1, ..., x_n\}$ *is a transcendence base*
- K/k *satisfies Maclane's Criterion*

Then for some x_i the set $\{x_1, ..., \hat{x_i}, ..., x_{n+1}\}$ is a ***separating transcendence base*** *for K/k.*

Note when $n = 0$ this means precisely that a simple algebraic extension satisfying Maclane's Criterion is separable.

Proof. By assumption x_{n+1} is algebraic over $k(x_1, ..., x_n)$, and therefore there exists a non-zero $F \in k[X_1, ..., X_{n+1}]$ such that $F(x_1, ..., x_{n+1}) = 0$. Let F be such a polynomial of minimal total degree (total degree = the maximum degree of a monomial with non-zero coefficient appearing in F). We claim it is irreducible. For suppose not, then one of the irreducible factors must vanish at $(x_1, ..., x_{n+1})$, and this factor would have smaller total degree.

We wish to show that not all powers of X_i appearing in F are multiples of p. Suppose this were the case then the monomials

$$\left\{x_1^{\alpha_1} ... x_{n+1}^{\alpha_{n+1}} \mid \lambda_\alpha \neq 0\right\}$$

are k-linearly dependent where λ_α are the coefficients of F and $\alpha \in \mathbb{N}^{n+1}$. Then by Maclane's Criterion the set

$$\left\{x_1^{\alpha_1/p} ... x_{n+1}^{\alpha_{n+1}/p} \;\middle|\; \lambda_\alpha \neq 0\right\}$$

is linearly dependent. This contradicts the minimality of F.

Choose X_i for which a non p-th power appears in F and define

$$F_i(T) := F(x_1, ..., x_{i-1}, T, x_{i+1}, ..., x_{n+1}) \in k[S_i][T]$$

where $S_i := \{x_1, ..., \hat{x_i}, ..., x_{n+1}\}$. By assumption $F_i(T)$ is non-zero and clearly $F_i(x_i) = 0$ so that x_i is algebraic over $k(S_i)$, and therefore by (3.19.44) K is also algebraic over $k(S_i)$. By (3.19.141) S_i is a transcendence base for K and in particular algebraically independent. Therefore $k[S_i][T]$ may be identified with $k[X_1, ..., X_{n+1}]$ and we may conclude that $F_i(T)$ is irreducible. Further $k[S_i]$ is a UFD so by (3.17.44) $F_i(T)$ is irreducible as a polynomial in $k(S_i)[T]$. By construction $F_i(T) \notin k(S_i)[T^p]$ so by (3.19.91) F_i is separable. Therefore x_i is separable over $k(S_i)$, and $K/k(S_i)$ is separable by (3.19.80) as required. ∎

Proposition 3.19.148
Let K/k be a finitely-generated field extension which satisfies Maclane's Criterion (e.g. k is perfect). Then K/k is separably generated.

More precisely every generating set contains a separating transcendence base.

Proof. When $\text{char}(k) = 0$ the generating set contains a transcendence base by (3.19.139). Every subfield of K also has characteristic 0, so then we are done by (3.19.103). So we may only consider the positive characteristic case, $p > 1$.

Suppose $K = k(x_1, ..., x_n)$. By (3.19.138) there is a (possibly empty) subset which is a transcendence base, say $\{x_1, ..., x_d\}$ after renumbering. By (3.19.44) $[K : K'] < \infty$ where $K' = k(x_1, ..., x_d)$, and we may choose K' such that

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the degree of inseparability $[K : K']_i$ is minimal. If K/K' is separable then we are done. Otherwise we may assume by renumbering that $K'(x_{d+1})/K'$ is not separable (3.19.80) and therefore $[K'(x_{d+1}) : K']_i > 1$. Then by (3.19.147) there is an index $i \leq d+1$ such that $[K'(x_{d+1}) : K'']_i = 1$ where $K'' := k(x_1, ..., \hat{x}_i, ..., x_{d+1})$. By multiplicativity

$$[K : K']_i = [K : K'(x_{d+1})]_i [K'(x_{d+1}) : K']_i > [K : K'(x_{d+1})]_i [K'(x_{d+1}) : K'']_i = [K : K'']_i$$

which contradicts minimality. ■

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